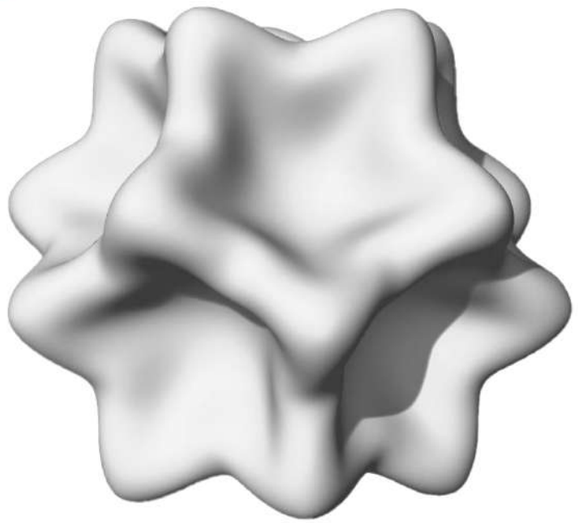
architecture thesis/

portfolio.



nithik v

"Mankind was born on Earth. It was never meant to die here"

- Cooper (Interstellar)

LUNA : THE MOON BASE

LIVING BEYOND EARTH

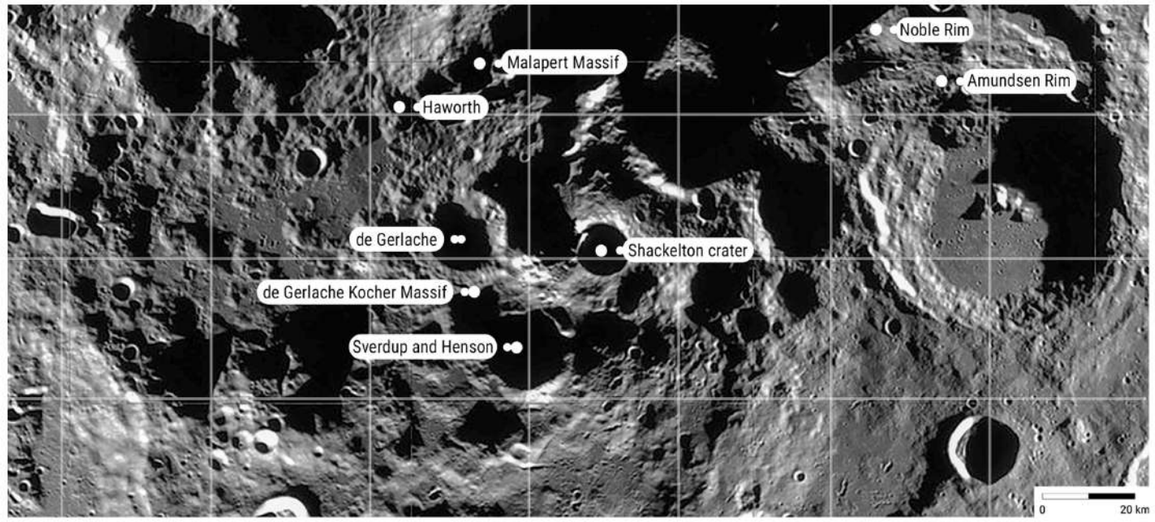
In the vast expanse of space, humanity's next frontier beckons: the lunar surface. As we envision a future where humans establish a sustained presence beyond Earth, the concept of lunar habitation emerges as a beacon of possibility. Within this cosmic endeavor lies a myriad of architectural challenges and opportunities, where design intersects with science, engineering, and human ingenuity. Crafting habitats to withstand the harsh lunar environment, while ensuring comfort, functionality, and sustainability, represents a pivotal chapter in extending human civilization's boundaries.



LUNAR SOUTH POLE

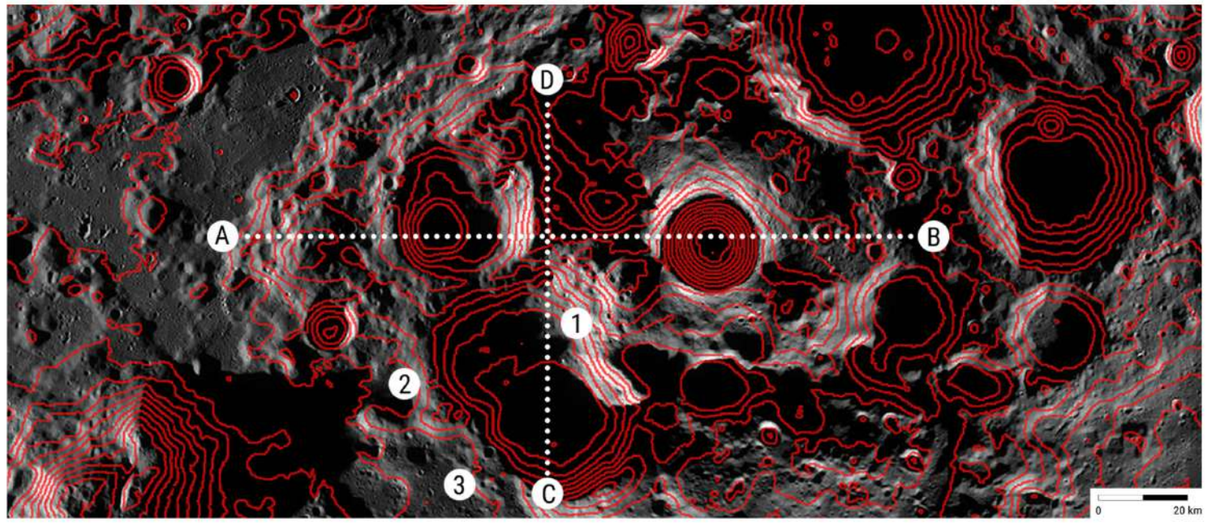
Lunar south pole is the suitable spot for habitation and future exploration. The permanently shadowed regions (PSR) on the Moon contain ice and other minerals as confirmed by Chandrayan, which is a vital resource for sustained presence on lunar surface. The rims of the crater in the south pole are illuminated for large periods of time which could be used to provide energy for the habitat and outpost. With the help of outpost scientists will be able to extract water and other volatile samples dating back to the formation of the Solar System.

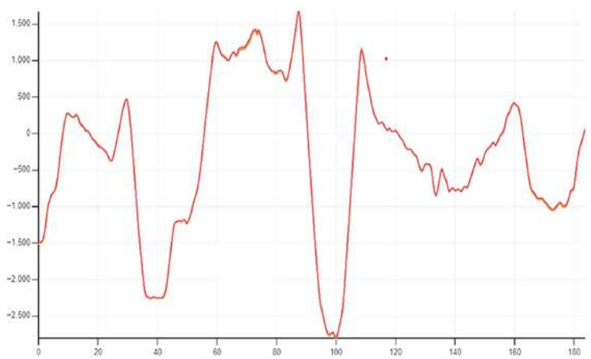
Site selection



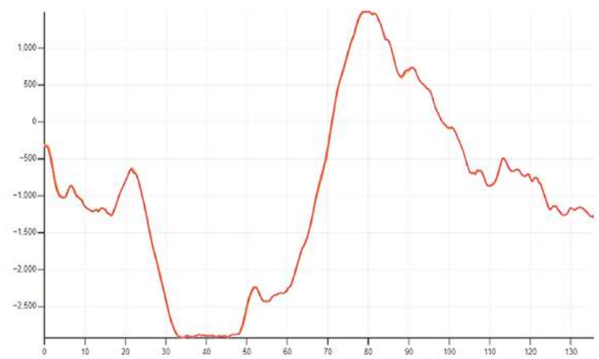
Key traits for selection:
Access to significant sunlight, which provides minimal temperature variations and potentially the only power source; continuous line-of-sight to Earth for mission support communications; mild grading and surface debris for safe landing and walking or roving mobility; and close proximity to permanently shadowed regions, some of which are believed to contain resources such as water ice.

Contour



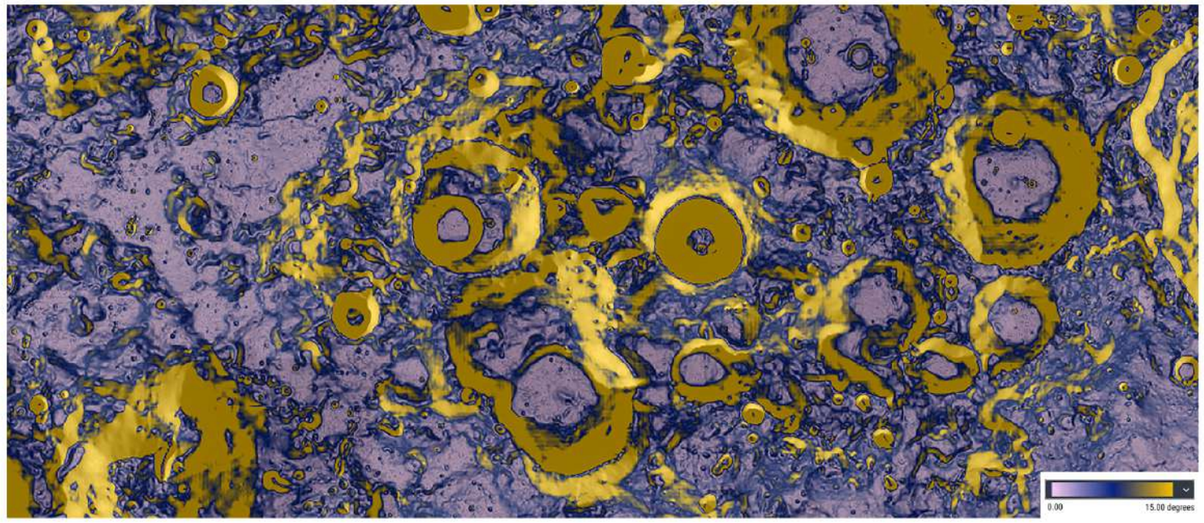


Elevation profile A to B



Elevation profile C to D

Slope



The slope of the terrain. Although the ideal base site would have a flat surface, even moderate slope angles may still be considered reasonably safe, depending on the task. A slope of 7 degree would allow spacecraft landing, while mobile surface operations (i.e., astronaut- only extra-vehicular activities [EVAs]) are rather safe on up to 15 degree slopes.

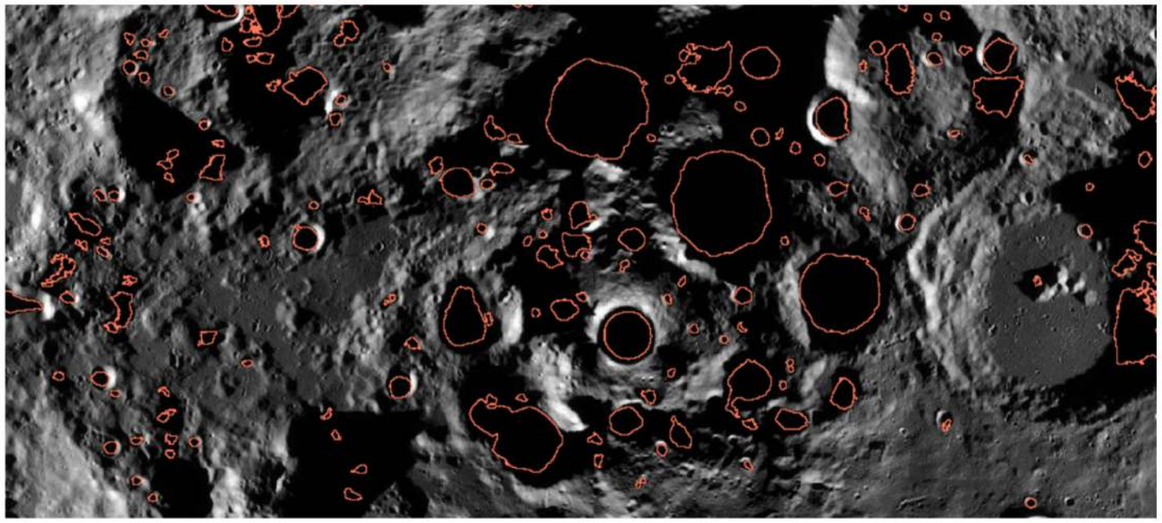


EVA (Extravehicular Activities and Transit) 0 - 15 degrees

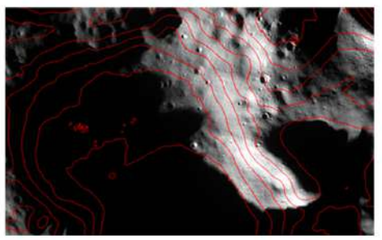


Landings and Take off 0 - 7 degrees

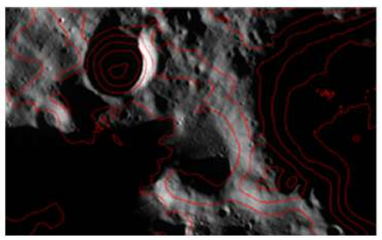
PSR



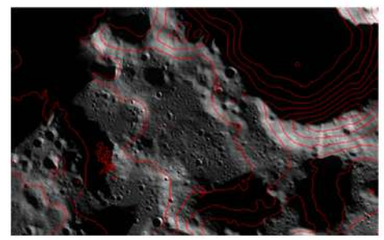
Permanently shadowed regions (PSRs) on the Moon are areas that receive little to no direct sunlight, causing these areas to remain incredibly cold and dark. These regions are typically found near the lunar poles, within craters or depressions that have a topography shielding them from direct sunlight. Designated area for base operations and expansion. The base site should have enough area for regular ground operations, say 81 km2, which includes part of the PSR, and the potential for future expansion. In these permanently shadowed regions (PSRs), temperatures remain low enough (<110 K) that volatile species such as water and CO2, delivered from a variety of sources, can be trapped and harbored for millions or billions of years.



1. Sverdup and Henson

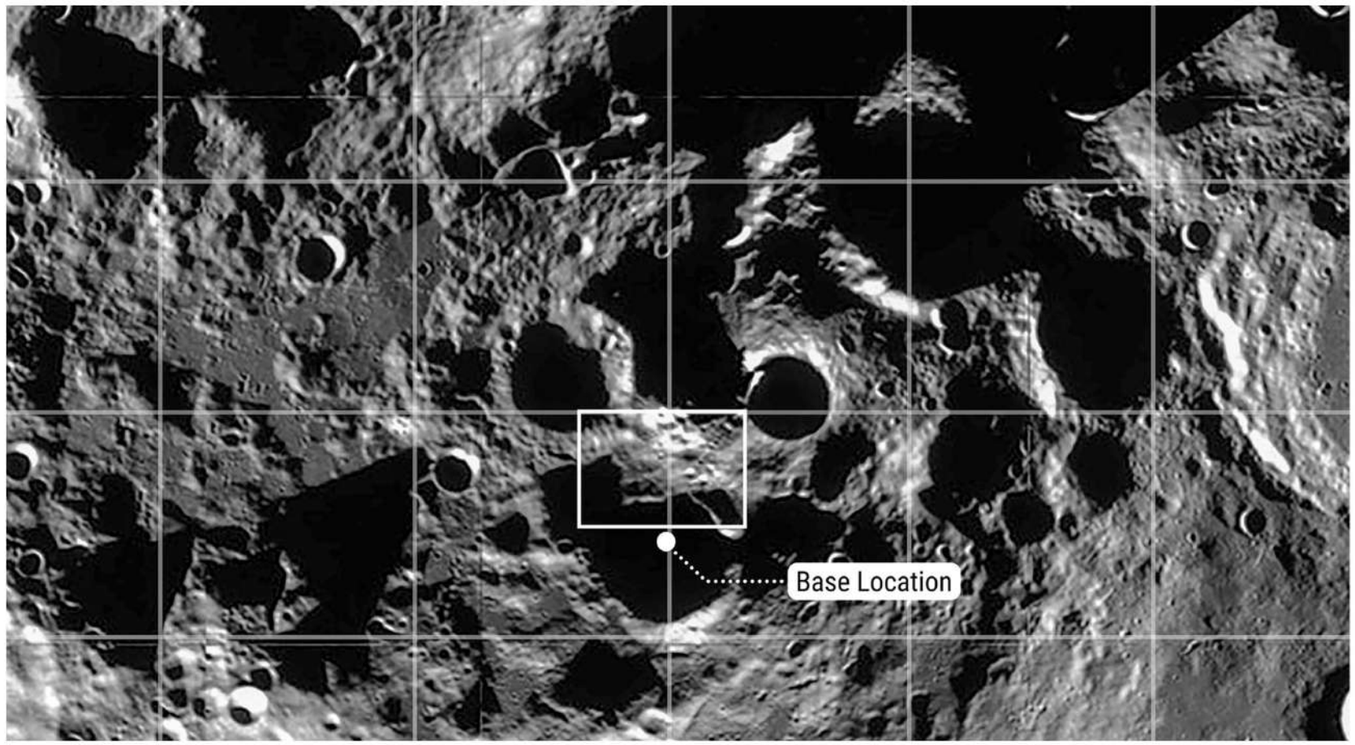


2. Bottom ridge - Sverdup and Henson

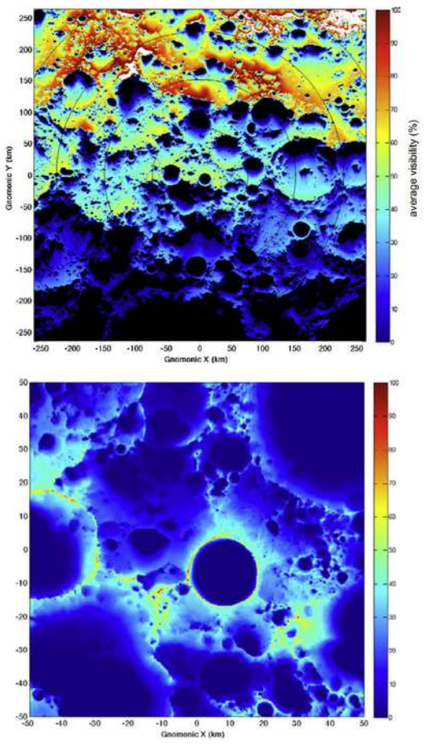


3. Base 30 km from - Sverdup and Henson

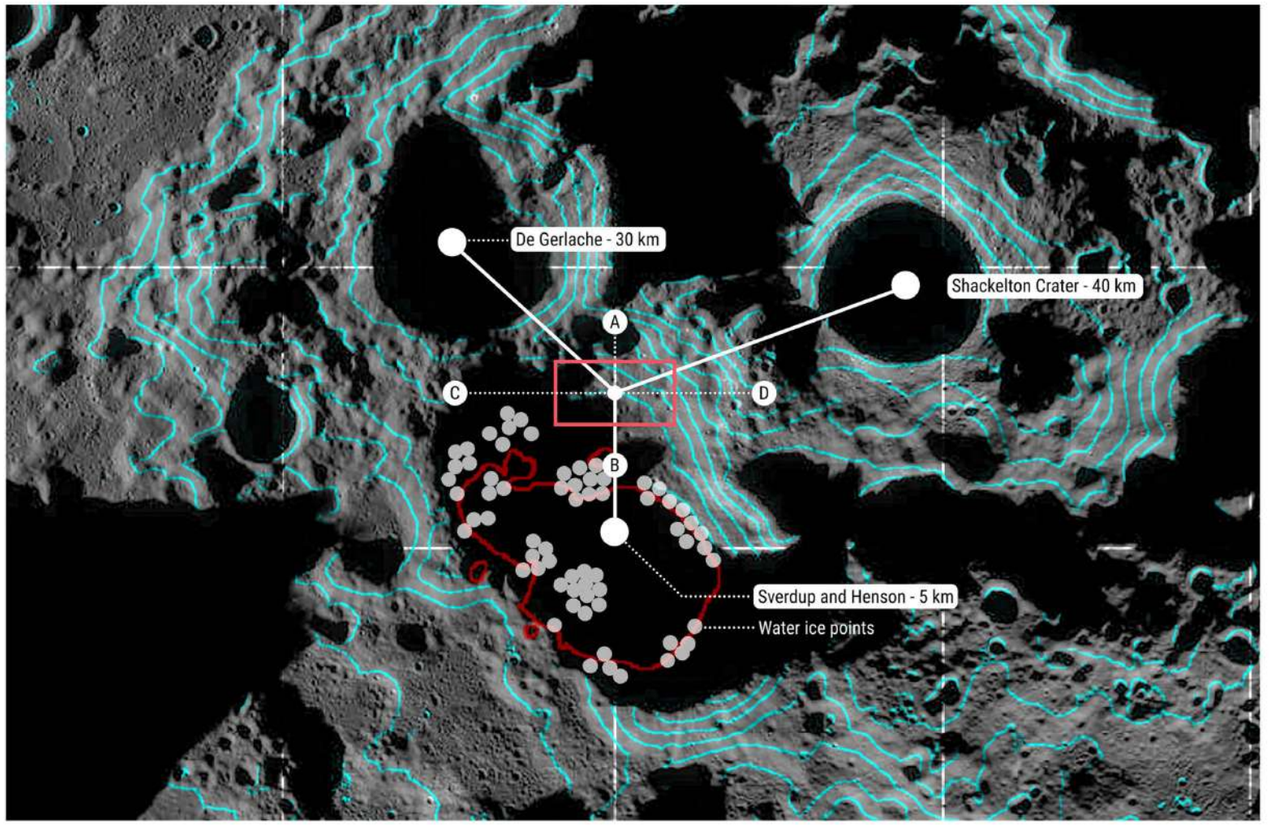
Base location

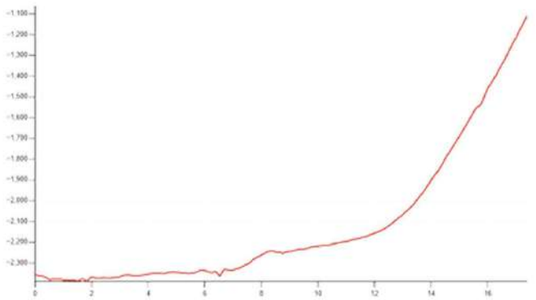


-88.77411 LAT, -128.90288 LON

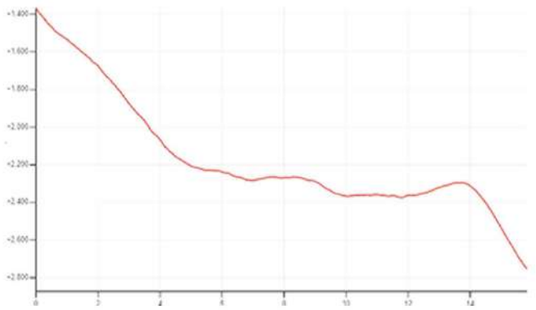


To set up a permanent human base, a place located within the Sverdrup-Henson crater is considered. Both crater floors exhibit PSRs with maximum temperatures of 110 K, which is important for direct access to water from the base. The base is located in the centre of four major craters in the lunar south pole and provides access to PSRs around the base. PSRs are never directly illuminated by sunlight and act as cold traps for ice and other volatiles. The rim between Henson and de Gerlache has illuminated areas for most of the lunar year. The nearest point to reach for water is the Sverdup and Henson crater wich has more water ice points than the shackelton crater. The base area is mostly flat and has the required amount of slope for construction, landing and transit which is between 0 to 15 degree.





Elevation profile A to B



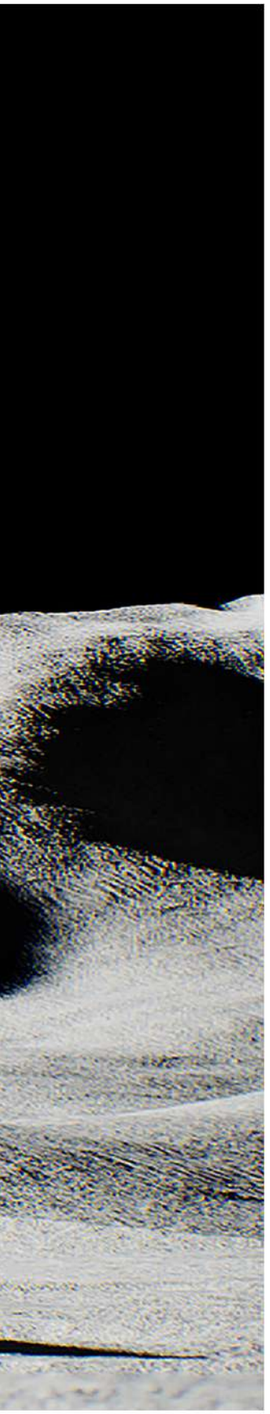
Elevation profile C to D

After using several selection criteria (abundance of water ice, the slope of terrain, usable energy sources, communications, and base expandability) to identify a suitable area for a future base in the southern polar crater Sverdrup-Henson. Due to the higher abundance of water ice, we found that the Sverdrup-Henson site is better suited to host a base than the nearby craters de Gerlache and Shackleton. The crater floor is partly in permanent shadow and exhibits numerous signatures of water ice.

STAGE 1

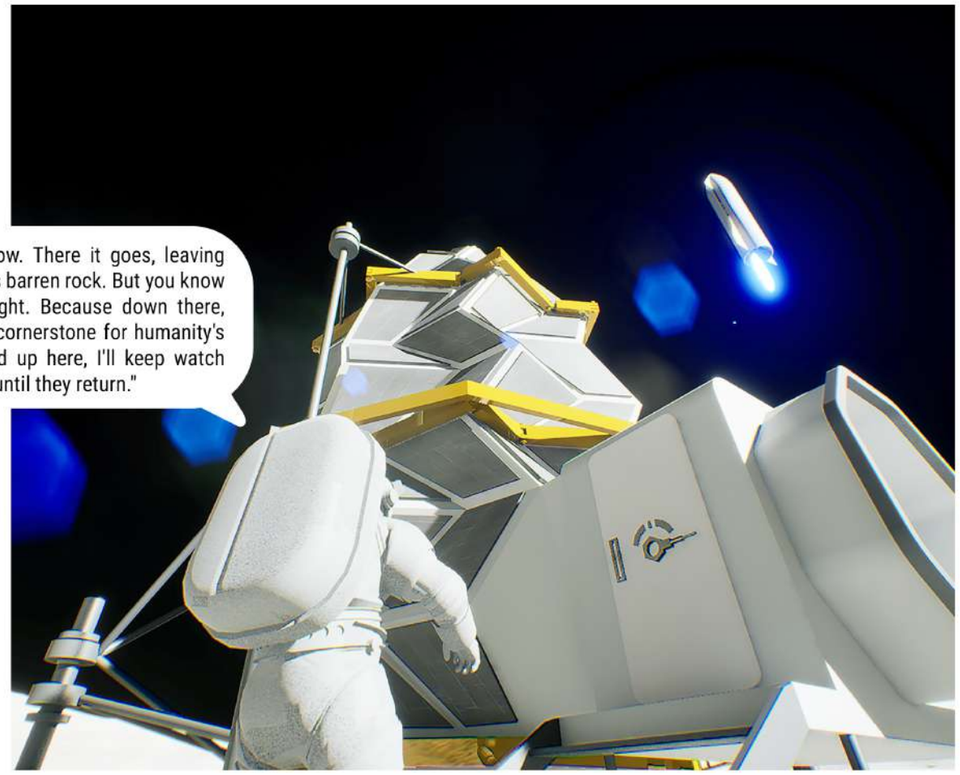
Habitat | Living module

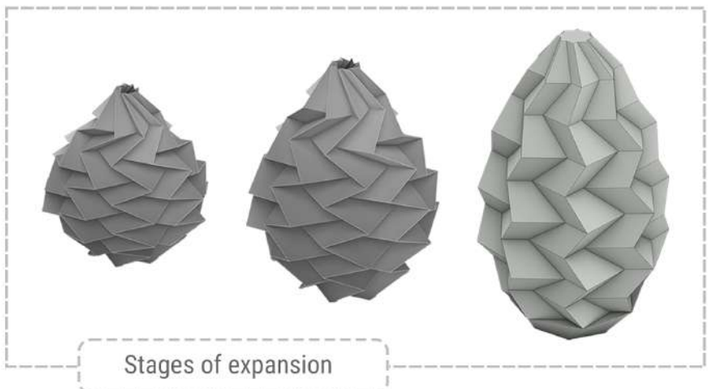
The initial step in establishing habitation involves deploying the living module onto the lunar surface, followed by ongoing research to facilitate further expansion. This module assembled on Earth using sustainable materials and pre-assembled, is then transported to the Moon under contracted arrangements to minimize payload and overall transportation costs. Upon arrival, it can be expanded to accommodate additional space requirements. There are two deployment options: the habitat can be delivered in its contracted form or fully expanded upon arrival on the lunar surface.





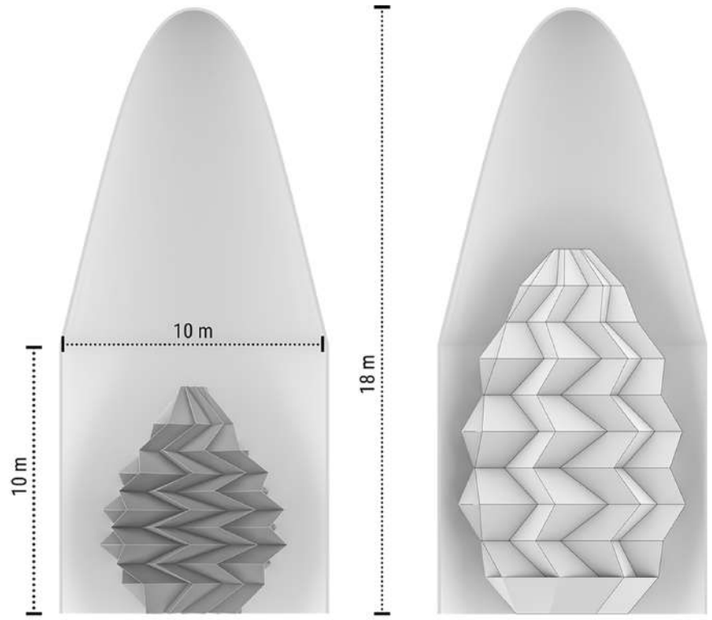
"Wow... just... wow. There it goes, leaving me behind on this barren rock. But you know what? That's alright. Because down there, I've just laid the cornerstone for humanity's next chapter. And up here, I'll keep watch over our dreams until they return."



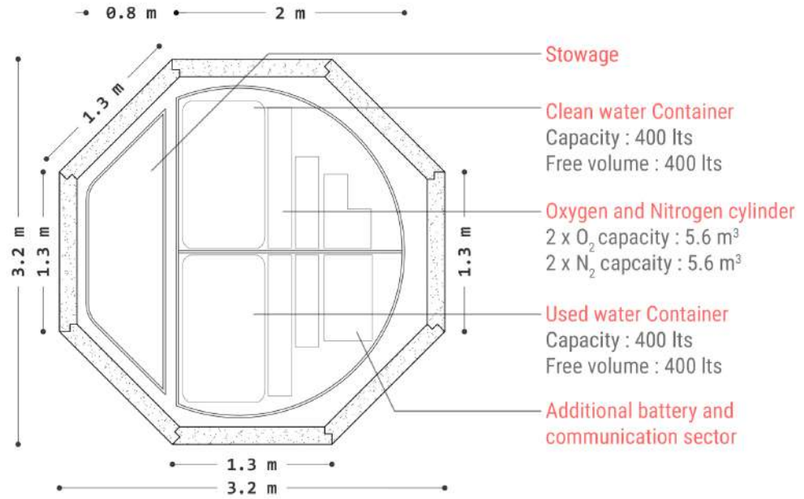


Stages of expansion

Miura origami offers the potential to create structures that, when expanded, can increase in volume to establish habitable environments on the lunar surface. This innovative approach significantly reduces launch costs and payload per transit from Earth to the Moon, paving the way for sustainable lunar habitation.

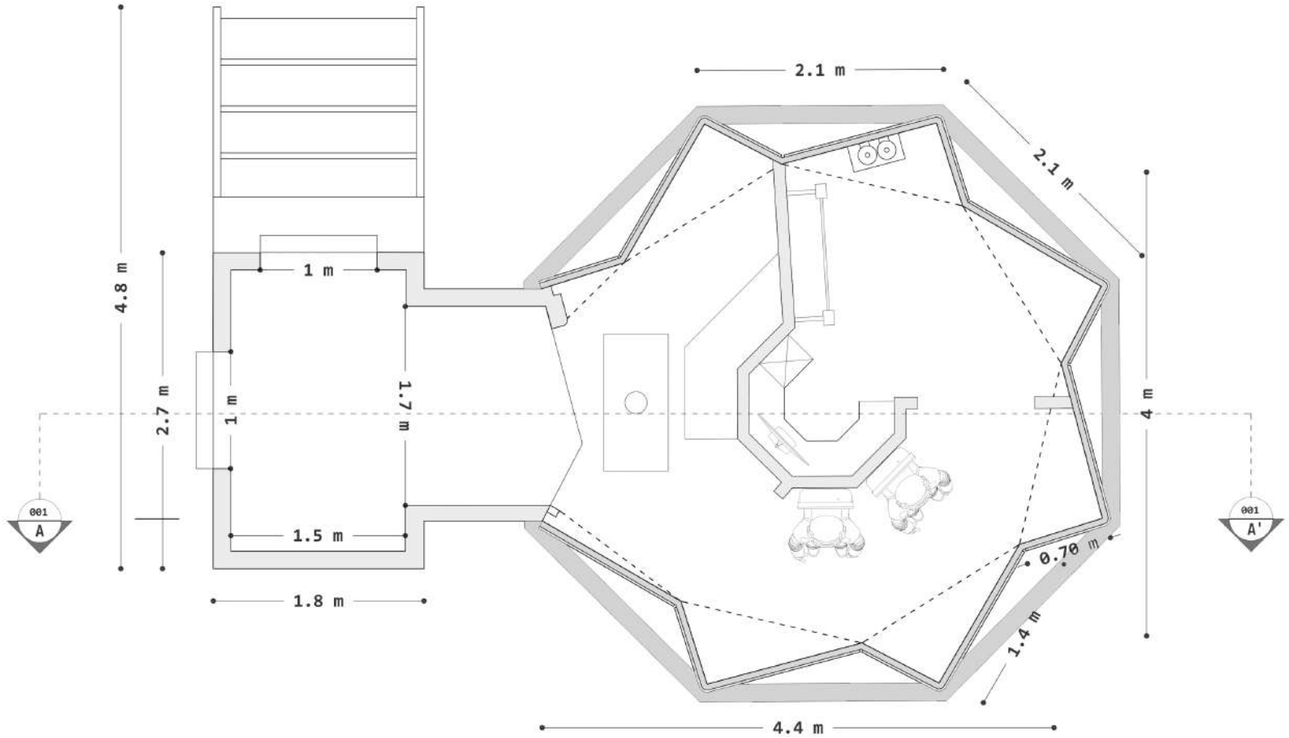


Note : The fairing size used are of standards from SPACE X's starship.

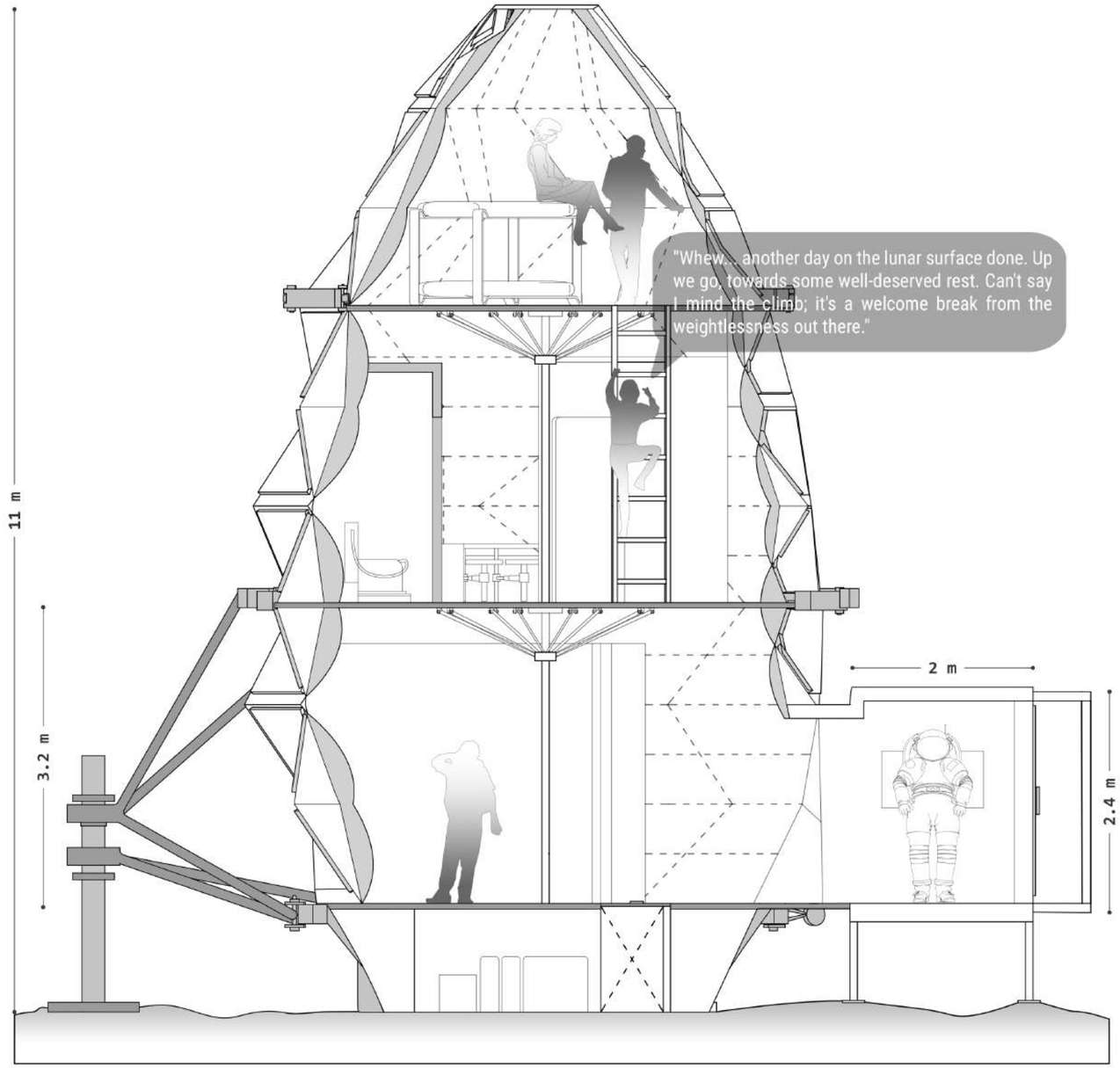


BASE PLAN

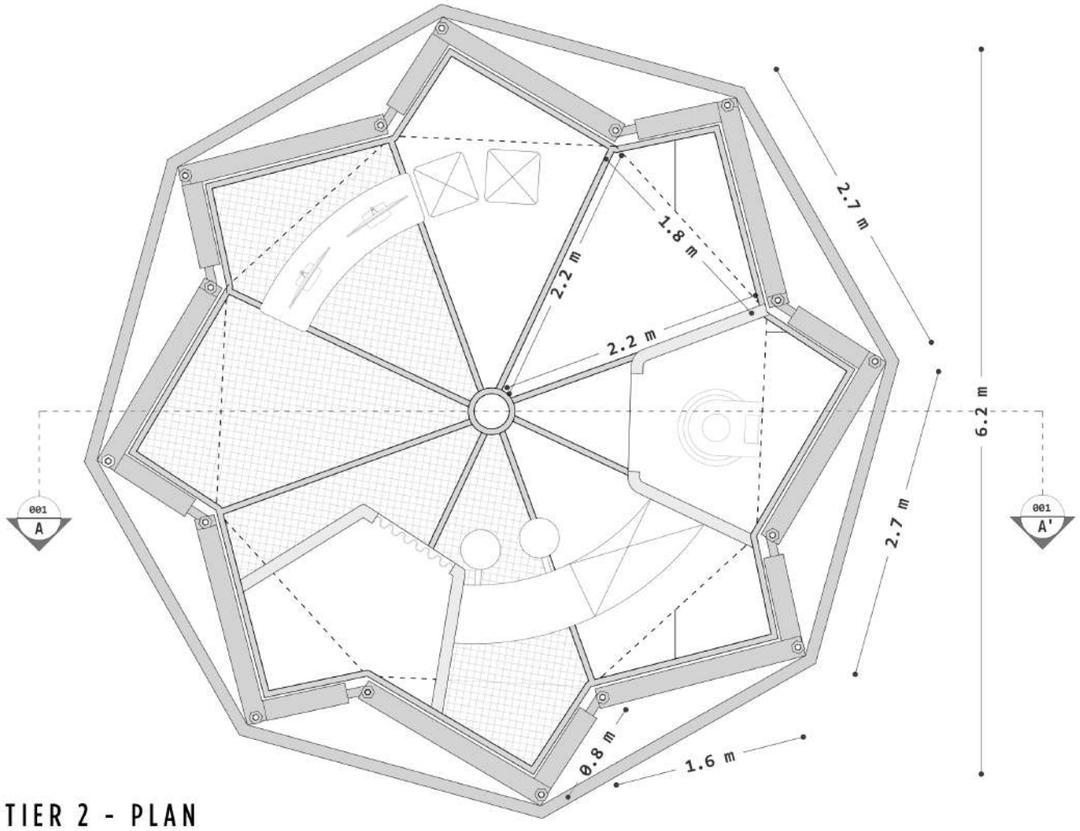
The living habitat comprises three tiers, each equipped with support systems pre-assembled in the base, along with additional space for storage. Tier 1 serves as the transition space between the hatch/airlock and the main living area. Within Tier 1, three zones are delineated: the transition zone, suit hangar, and ARED with ECLSS control system. Partitions are assembled once the habitat is expanded and the floor plate supports are fully deployed. The ARED device is utilized for resistive exercise, a mandate by NASA requiring a minimum of 2 hours of exercise. The ECLSS control system oversees oxygen levels and other environmental parameters within the habitat.



TIER 1 - PLAN

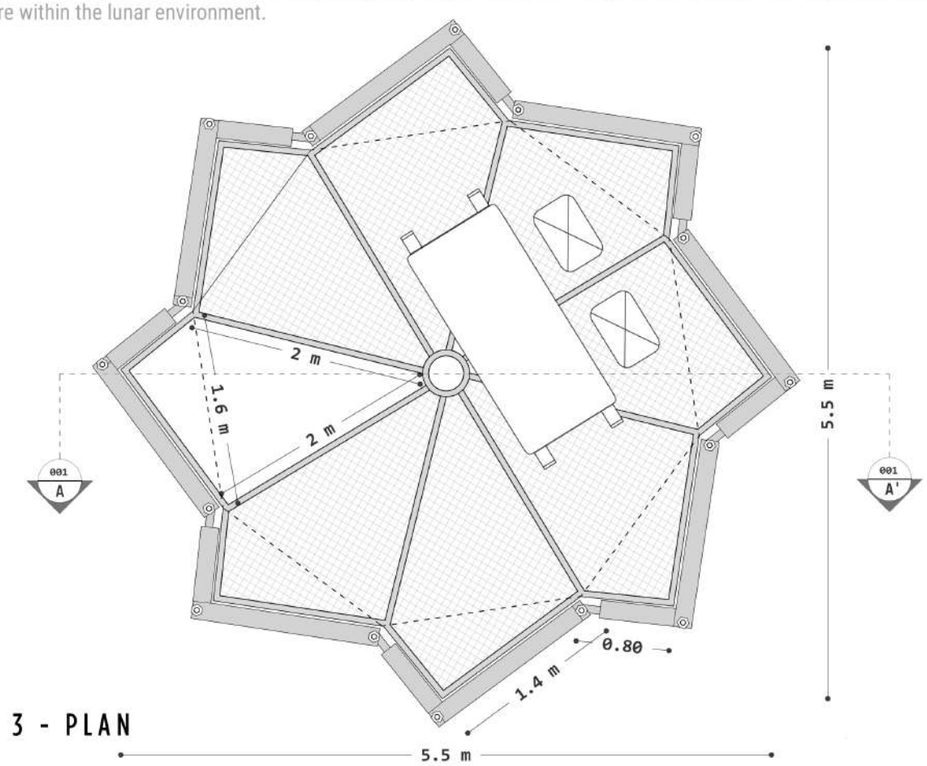


SECTION AA'
SCALE: 1:50 001

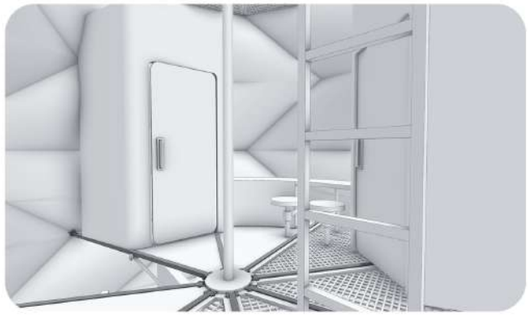


TIER 2 - PLAN

The radial layout of spaces and accessories in Tier 2 enhances circulation space at the center. As we ascend through the tier, the radius of the floor plates decreases, consequently reducing the cantilever angle of the module. This adjustment aids in managing pressure within the lunar environment.

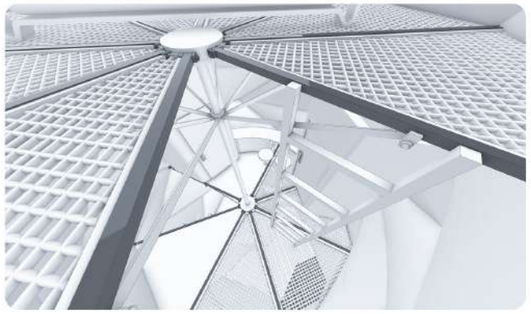


TIER 3 - PLAN



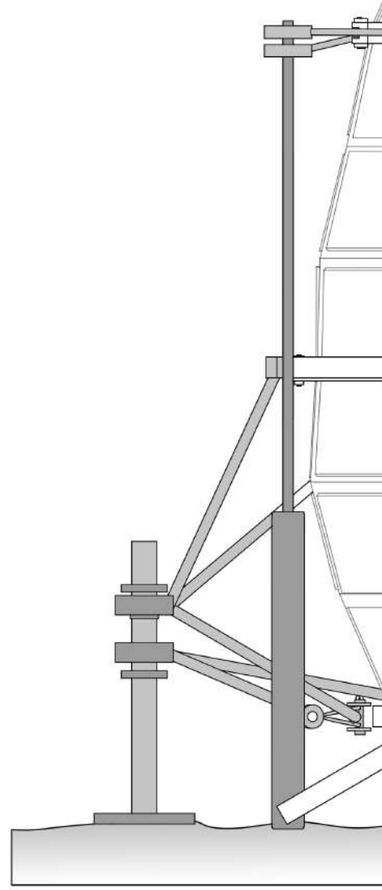


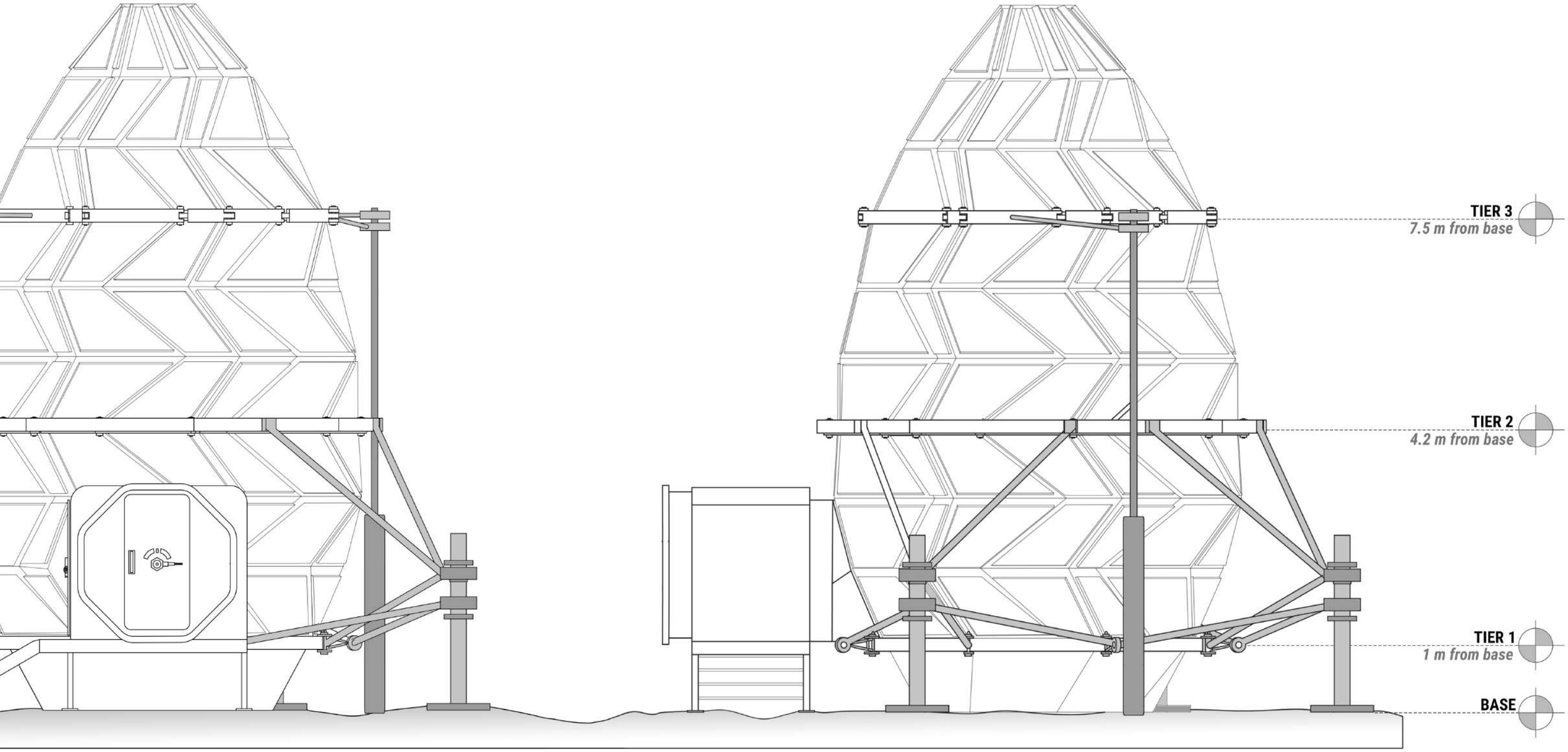
Tier 2 encompasses all essential areas for hygiene, dining, medical storage, and additional stowage. Chairs can be pulled out as needed to maximize usable space, with the spaces arranged radially from the center.





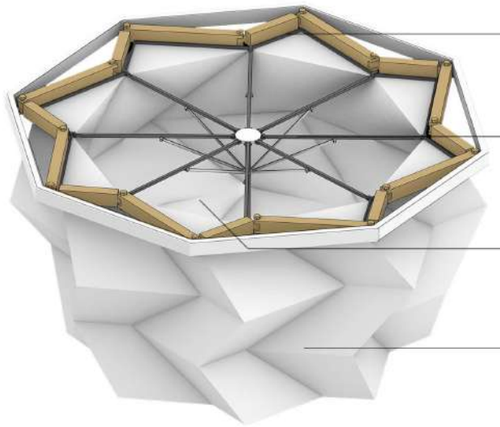
Tier 3 is designated for personal use, offering sleeping quarters and personal storage, while Tier 2 houses all other hygiene facilities and workstations. This serves as a secondary space where research and work can continue from the labs, functioning as the primary control center for module services.





ELEVATION A 001
SCALE: 1:50

ELEVATION B 001
SCALE: 1:50

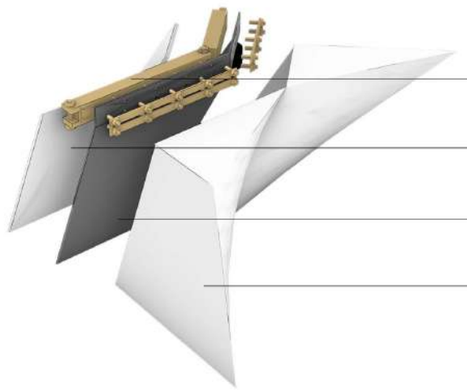


Frame (Collapsabe)
(Titanium/ Titanium alloy Ti-6Al-4V)

Umbrella support for floor plates
(Titanium/ Titanium alloy Ti-6Al-4V)

Interior layer
(Vaccum insulated panels)

Exterior layer
(Carbon Fibre Reinforced Polymer)



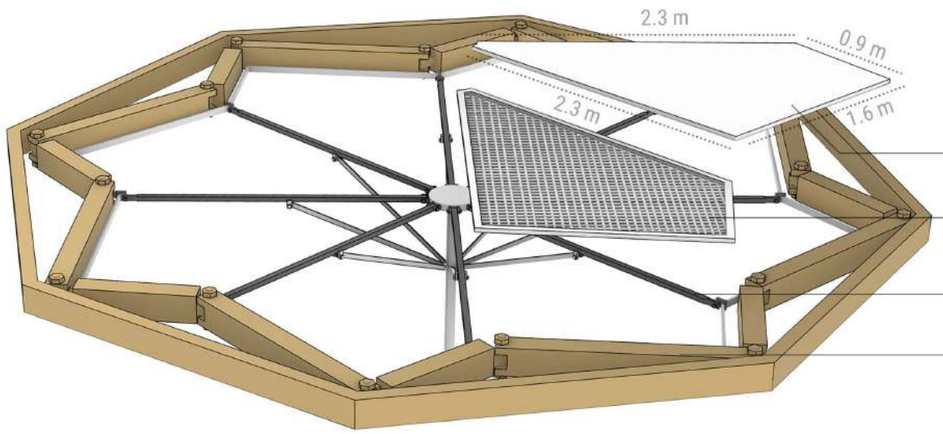
Frame
(Titanium/ Titanium alloy Ti-6Al-4V)

Exterior Panel
(Mylar - thermal blankets)

Exterior Layer
(Carbon Fiber Reinforced Polymers)

Interior Layer
(Vaccum insulation panels)

PANEL DETAILS



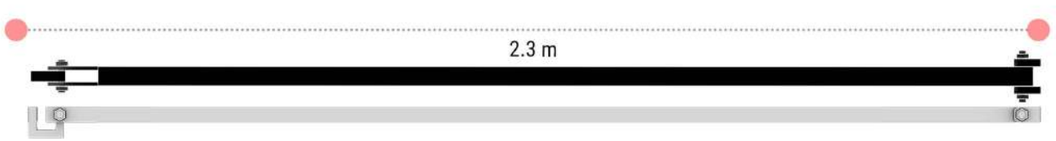
Floor plates
Solid plate (CFRP)

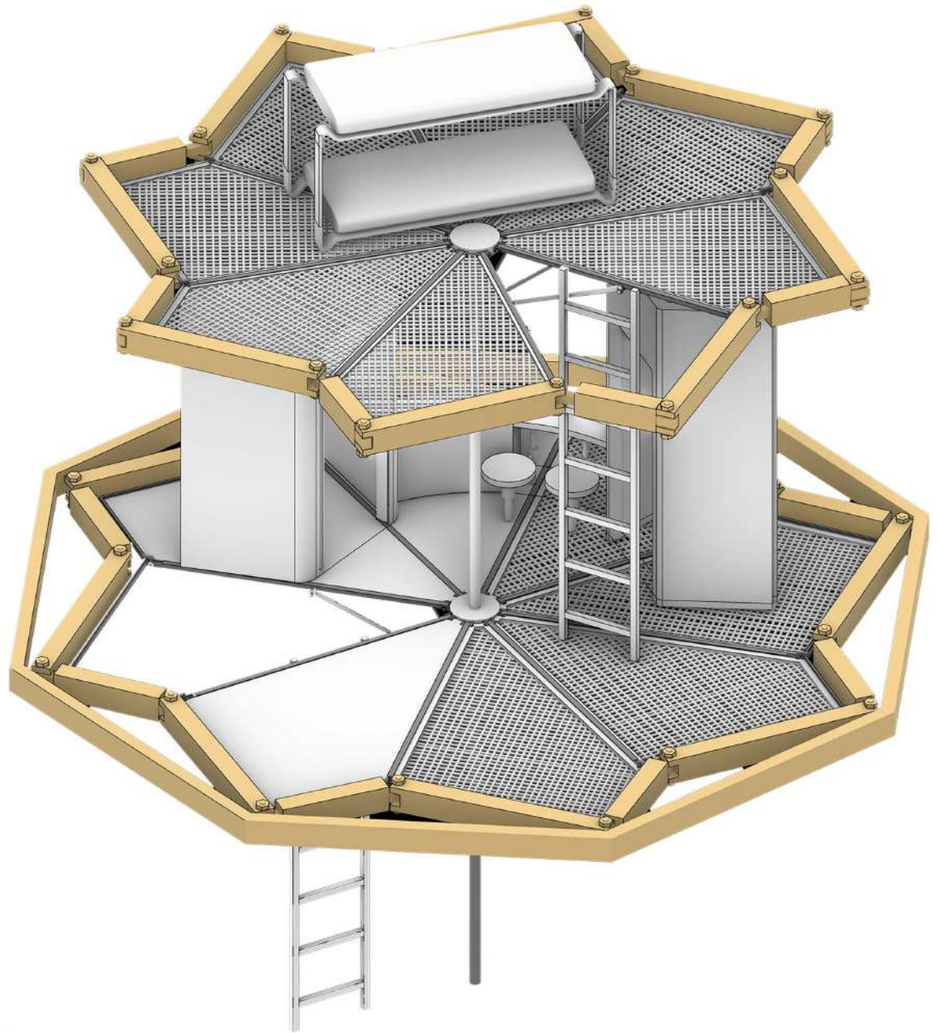
Floor plates
Aluminium grill floor

Umbrella support
floor plates

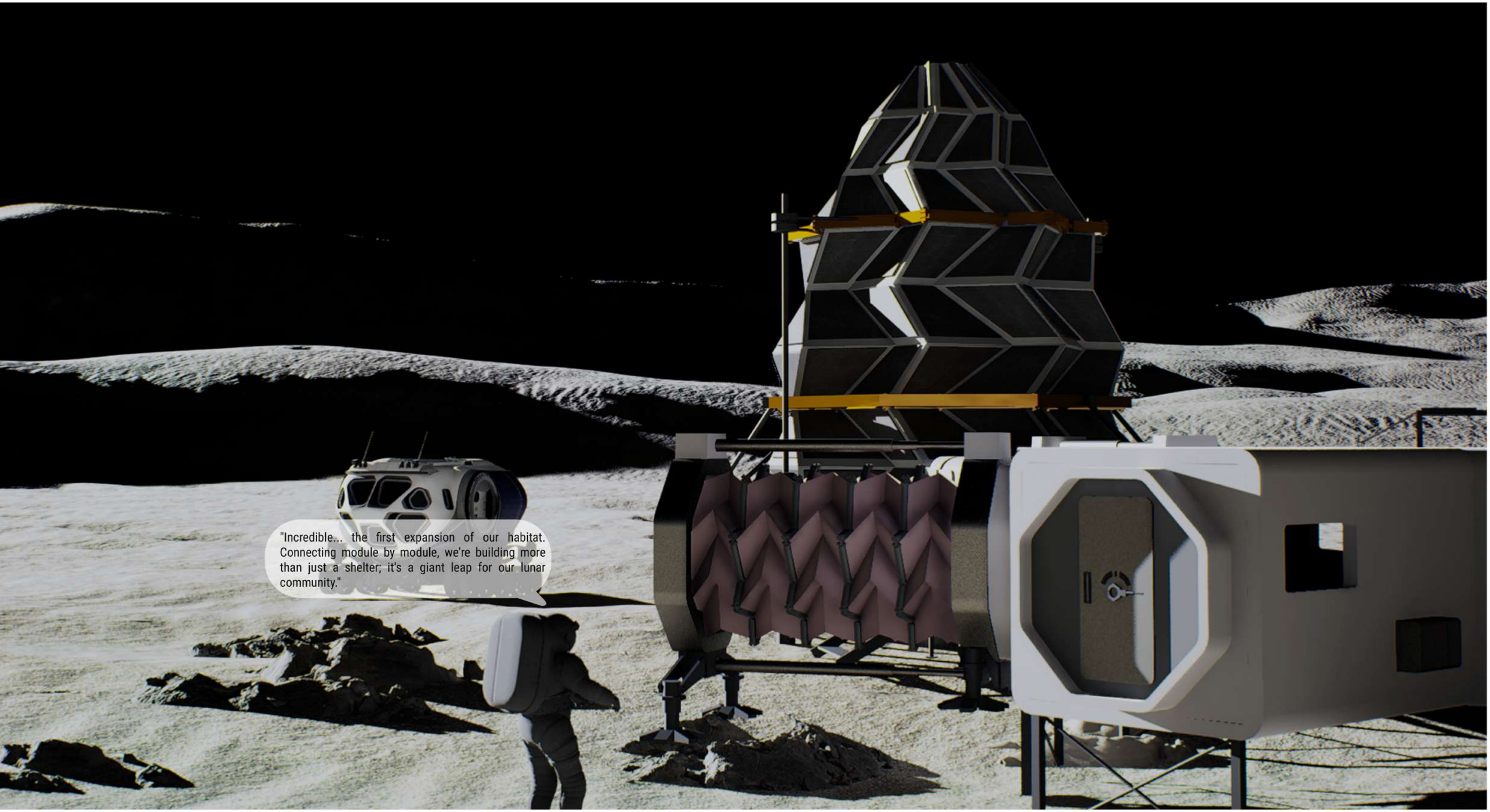
Frame (Collapsabe)
(Titanium/ Titanium alloy Ti-6Al-4V)

The deployment of umbrella joints aids in supporting the floor plates. Two types of floor plates are utilized: solid plates made of CFRP are employed in areas where hygiene and kitchen facilities are arranged, while aluminum grill floors are used elsewhere. The Moon's gravity contributes to reducing the transfer of dead load.





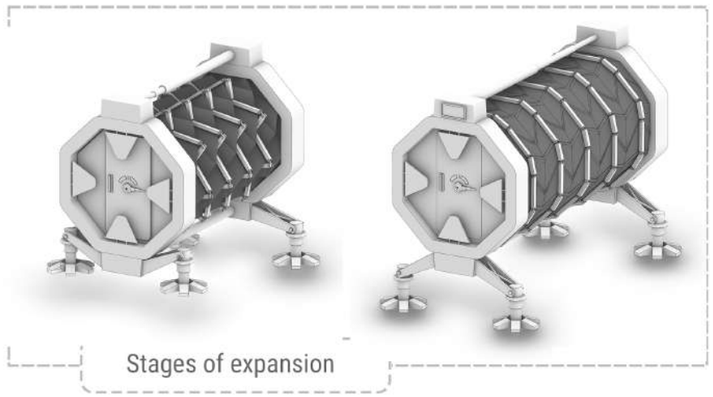




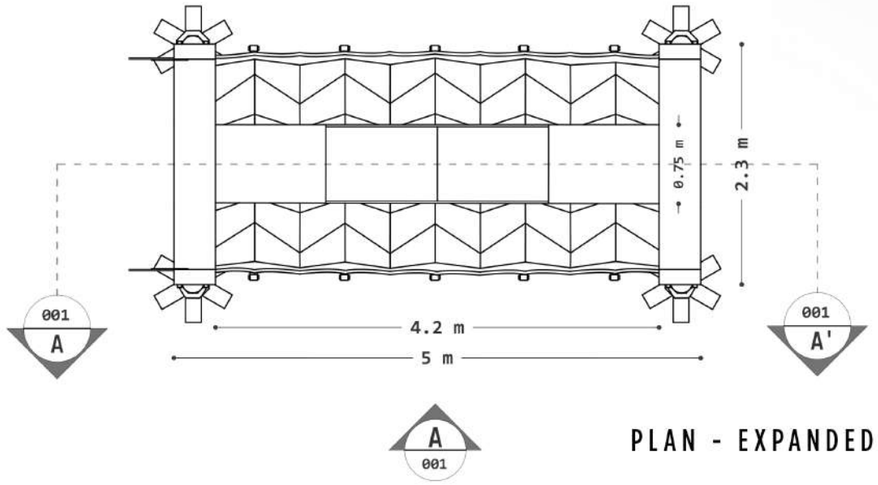
STAGE 2

Air tube | Connecting module

The second crucial phase in the development of lunar habitation involves the expansion of the habitat, a pivotal step facilitated by the integration of a connecting module or an air tube. This expansion mechanism serves as a vital conduit, facilitating smooth transitions not only between individual living modules but also towards communal spaces essential for fostering social cohesion within the settlement. The mechanism, powered by two electric telescopic actuators meticulously designed for lunar conditions, ensures the seamless expansion of the tube, adapting to the evolving needs of its inhabitants in their extraterrestrial lifestyle.

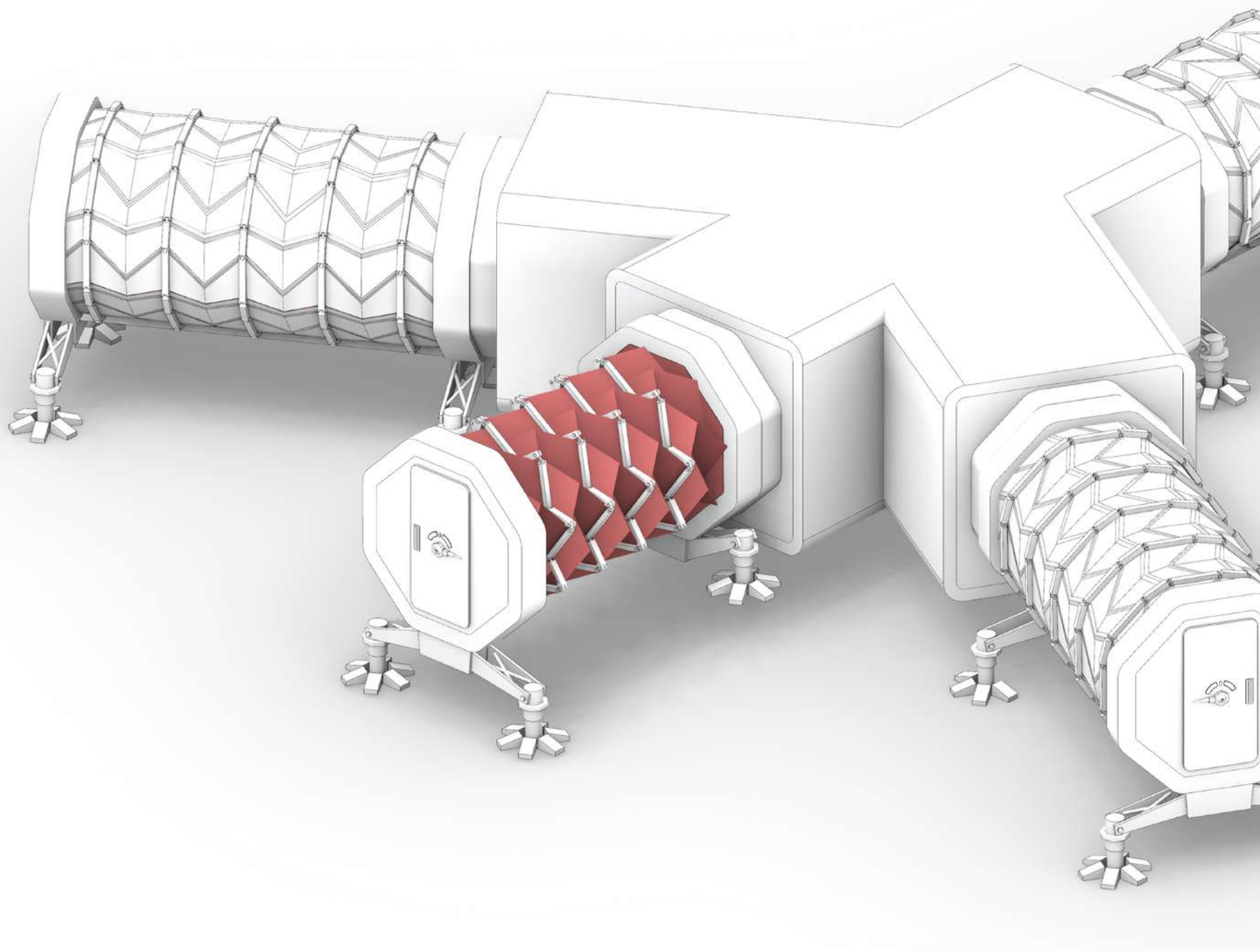


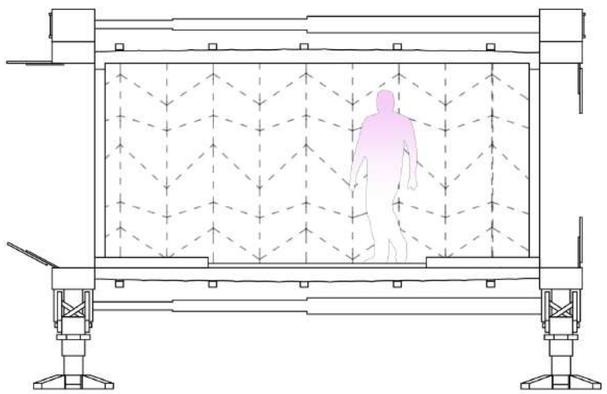
Stages of expansion

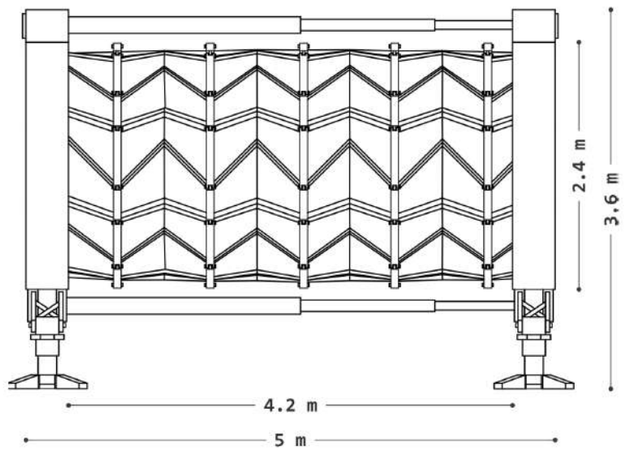


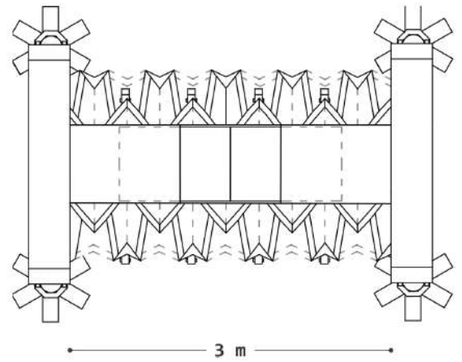
A
001

PLAN - EXPANDED

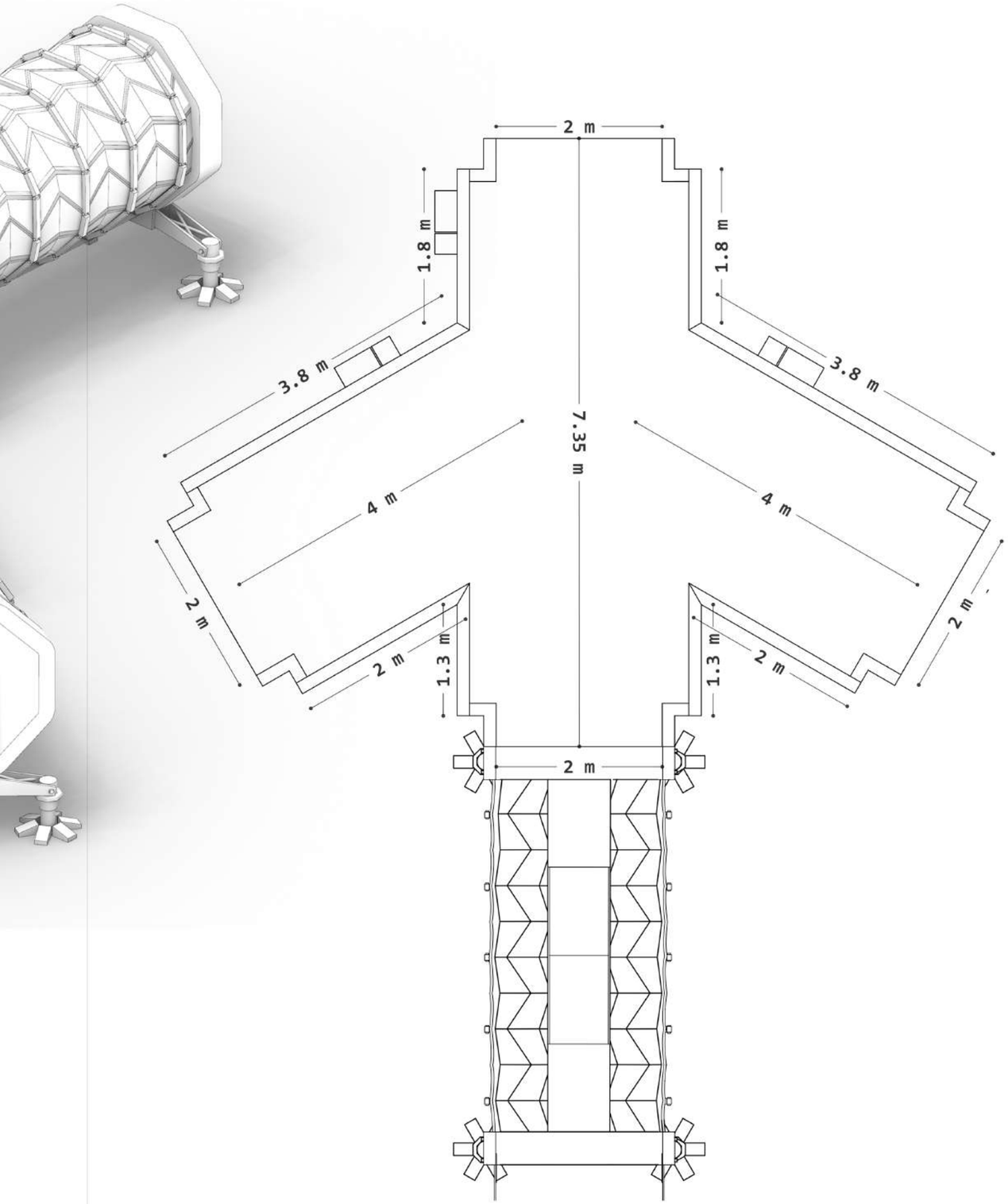


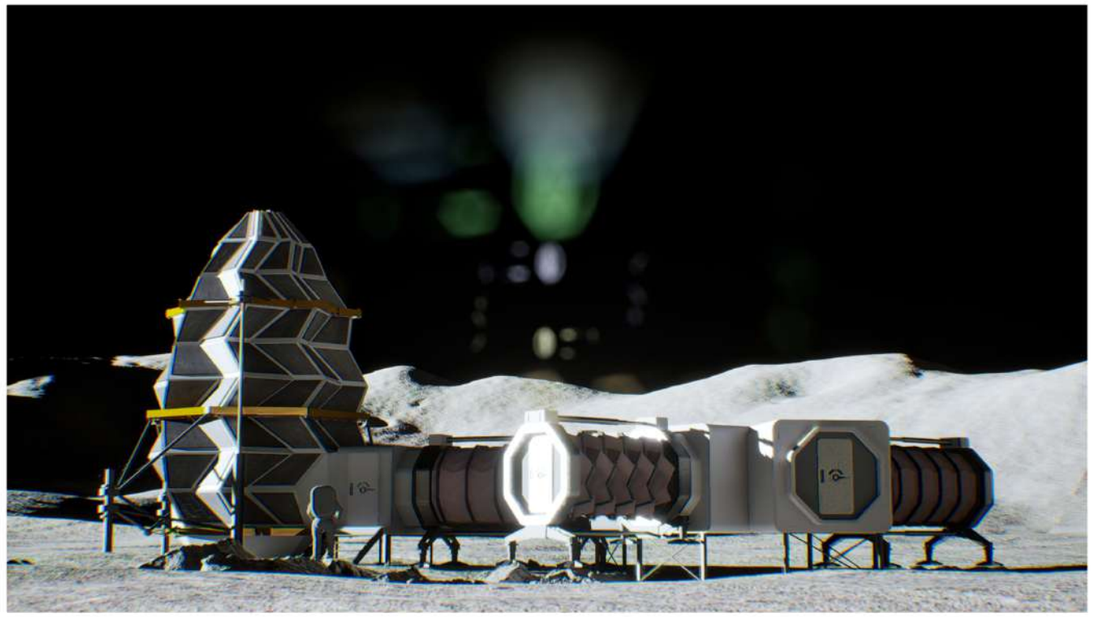


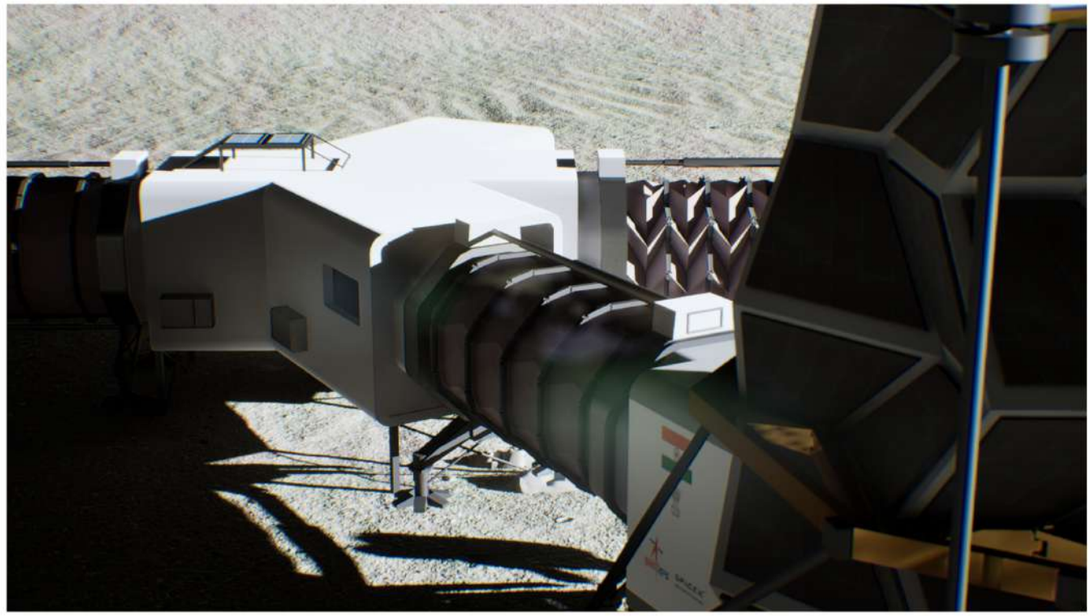




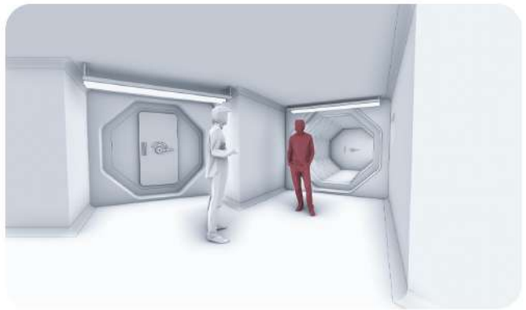
PLAN - COMPRESSED











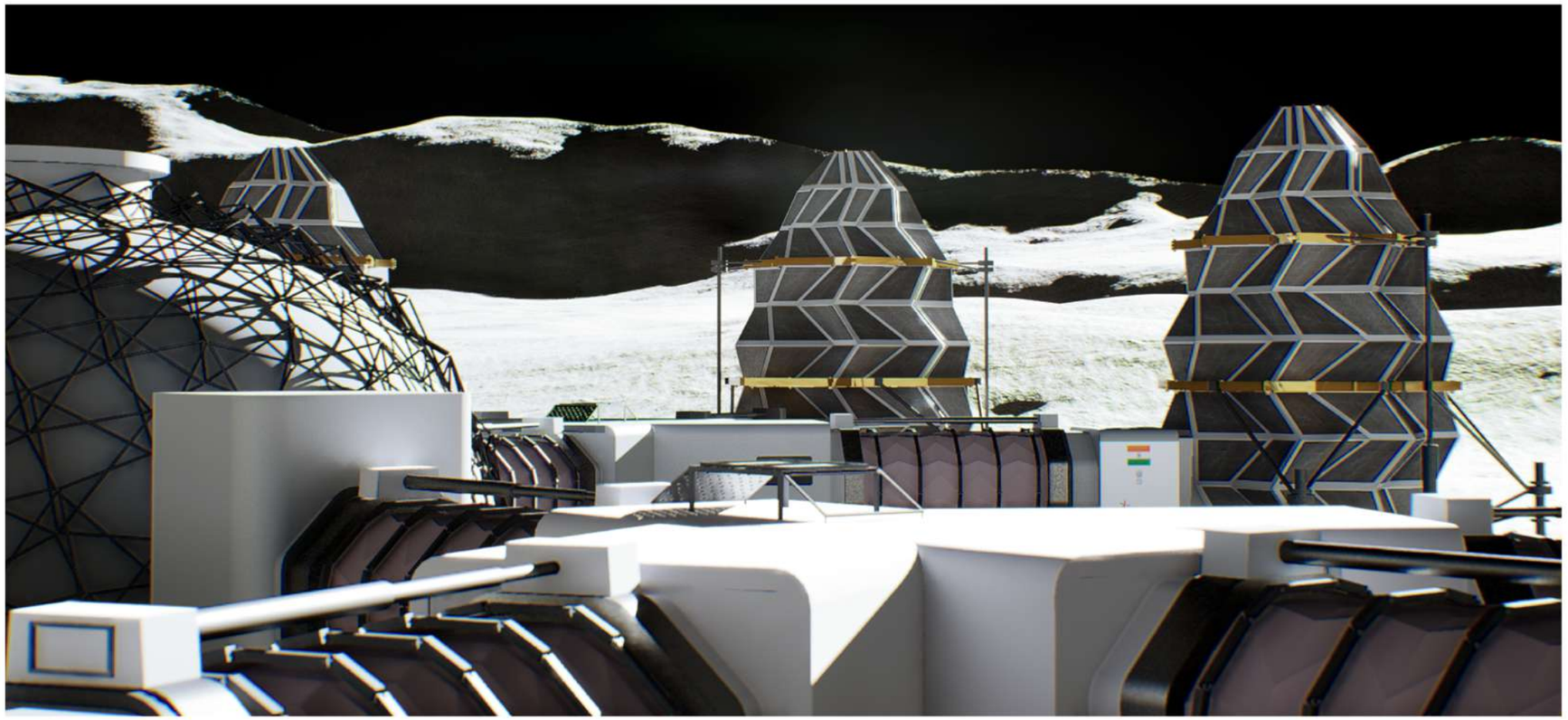
This area serves as the primary transit zone connecting various functions on the lunar surface, while also providing a small overflow space. Telescopic floor plates facilitate expansion and support the structure of the transit tube, serving as a hub between modules.

STAGE 3

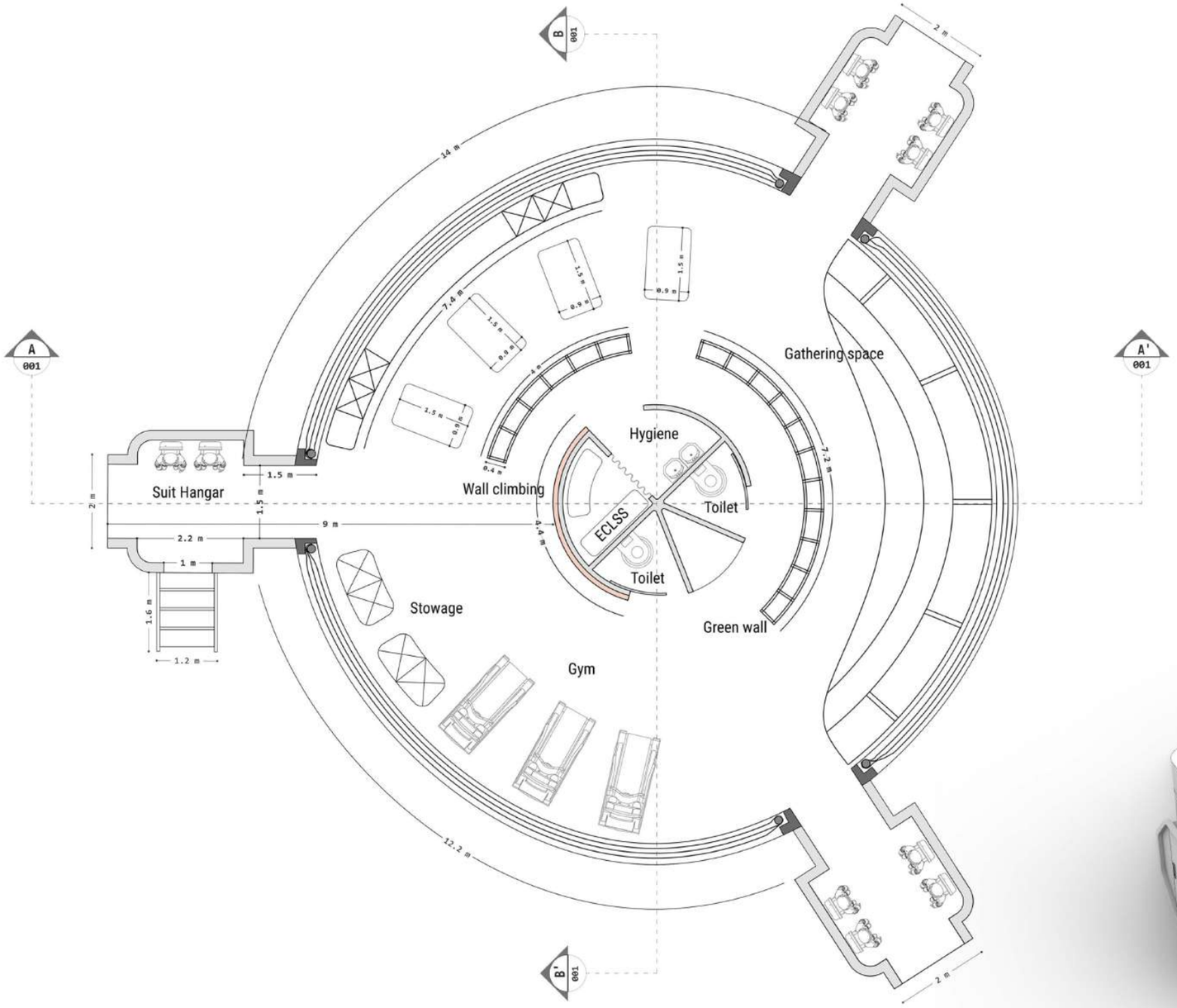
Elysium | Communal space

The third phase entails constructing communal spaces on the lunar surface, serving as pivotal nodes between living habitats and laboratories. Positioned amidst diverse functions, these spaces offer public gathering areas on lunar soil, aimed at exploring the potential impacts on mental well-being. This alternative presents another viable option for establishing settlements on the lunar surface. Mechanical joints offer support and maintain pressure within the inflatable shell, thereby reducing payload and transportation costs. Once deployed, the communal space is anchored to the 3D-printed base.

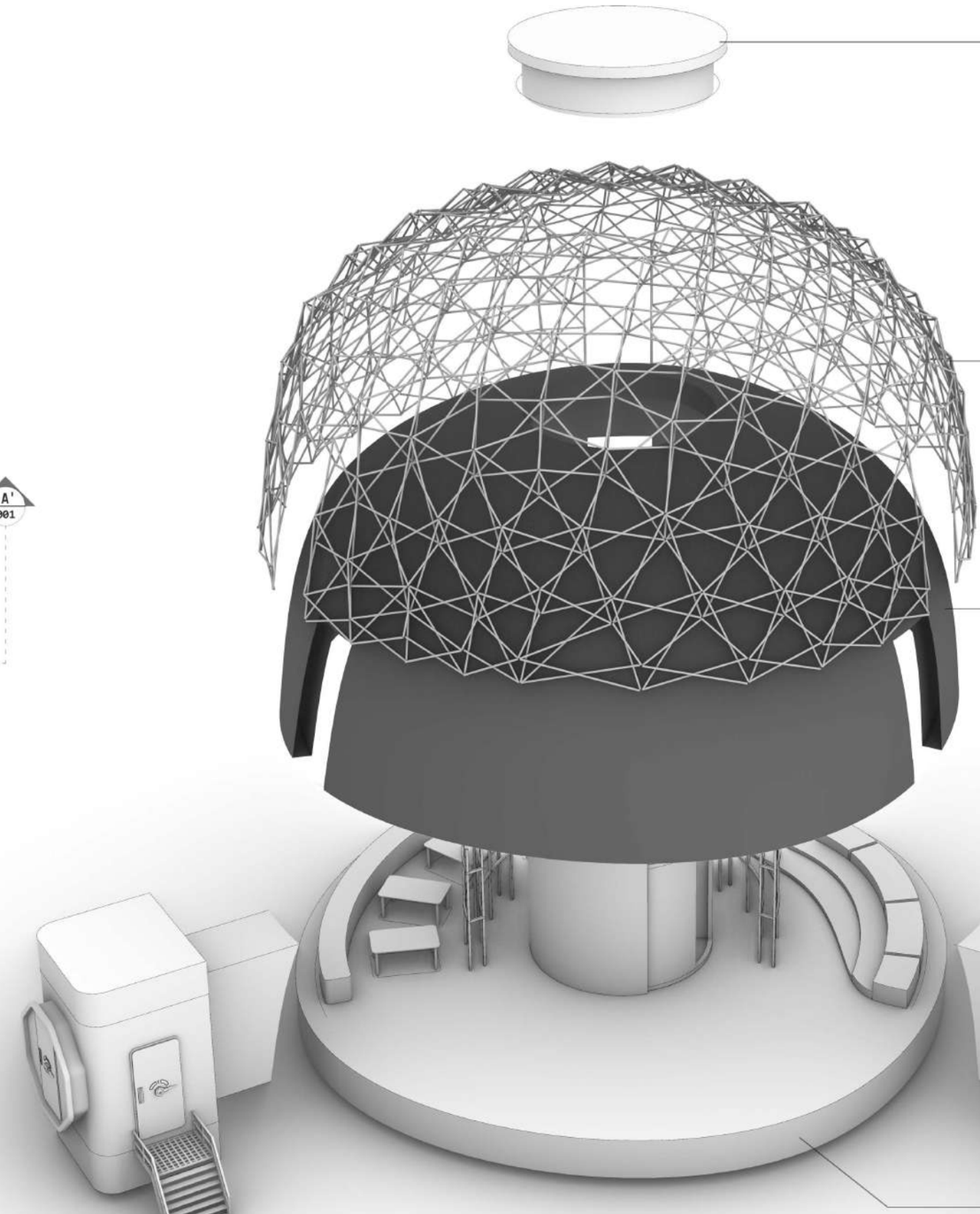




The focus shifts to building communal areas on the lunar surface, acting as crucial connections between living habitats and laboratories. Positioned strategically within various activities, these spaces provide public gathering spots on lunar terrain, aiming to study their potential effects on mental health. This presents an additional viable approach to forming settlements on the lunar surface. Mechanical joints provide structural support and regulate pressure within the inflatable shell, thus lowering payload and transportation expenses. After deployment, the communal space securely fastens to the 3D-printed base.



FLOOR PLAN - COMMUNAL HUB

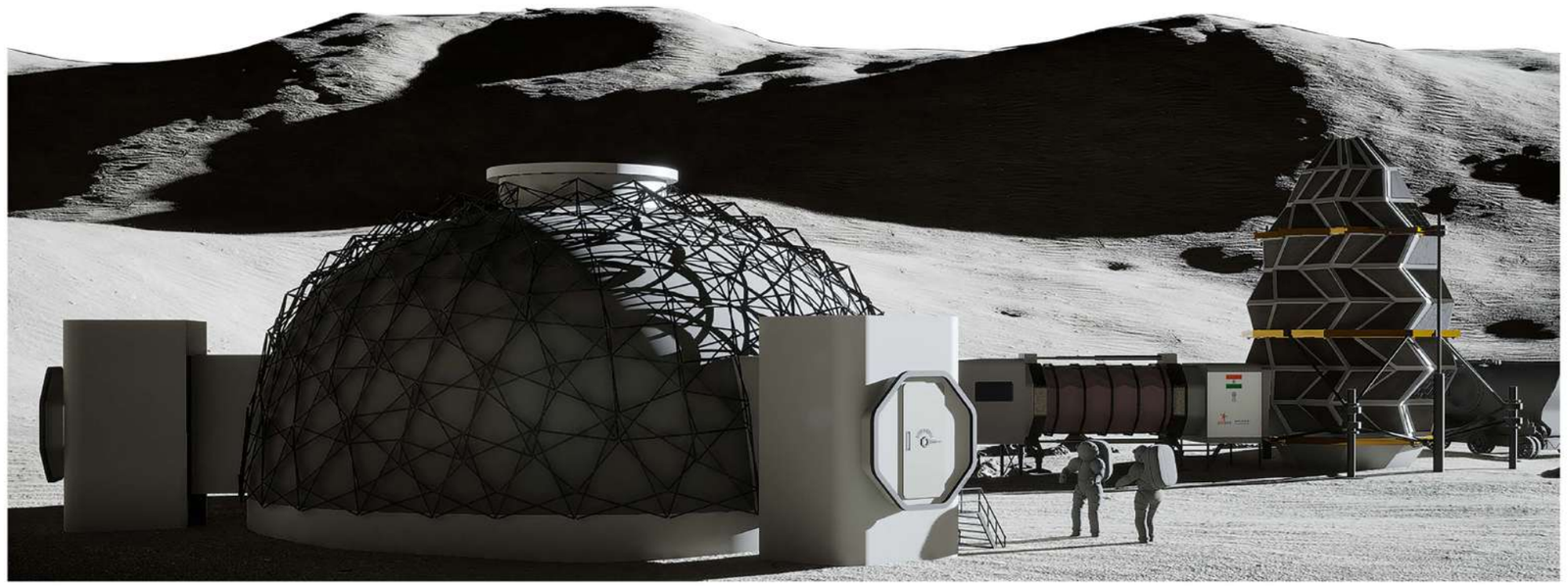


EXPLODED VIEW

Roof shell
Weight placed to hold the pressure inside the shell

Outer shell
Deployable support skeleton made with titanium linkages

Inner Shell
Inflatable material made up of 4 layers

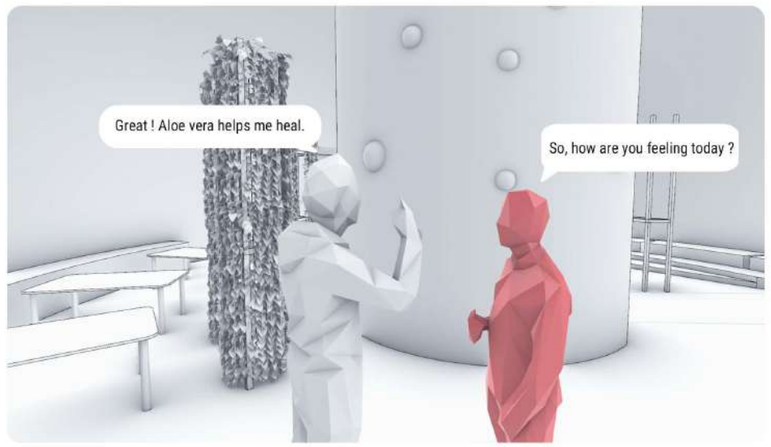


Prefab modules
Hatch/airlock prefabricated and installed on the surface before shell deployment



Base
3d printed base layer on leveled lunar surface





How do we implement our cultural values ?

As humans, we aspire to infuse our cultural values into new frontiers we explore. This shared space serves as a means to enhance the bond among inhabitants of the lunar base. The selection of plants for the green wall is based on their origins and common usage across multiple continents. Integrating these plants into the living areas of the habitat will contribute to preserving mental health and alleviating stress and anxiety often experienced in confined environments.

STAGE 4

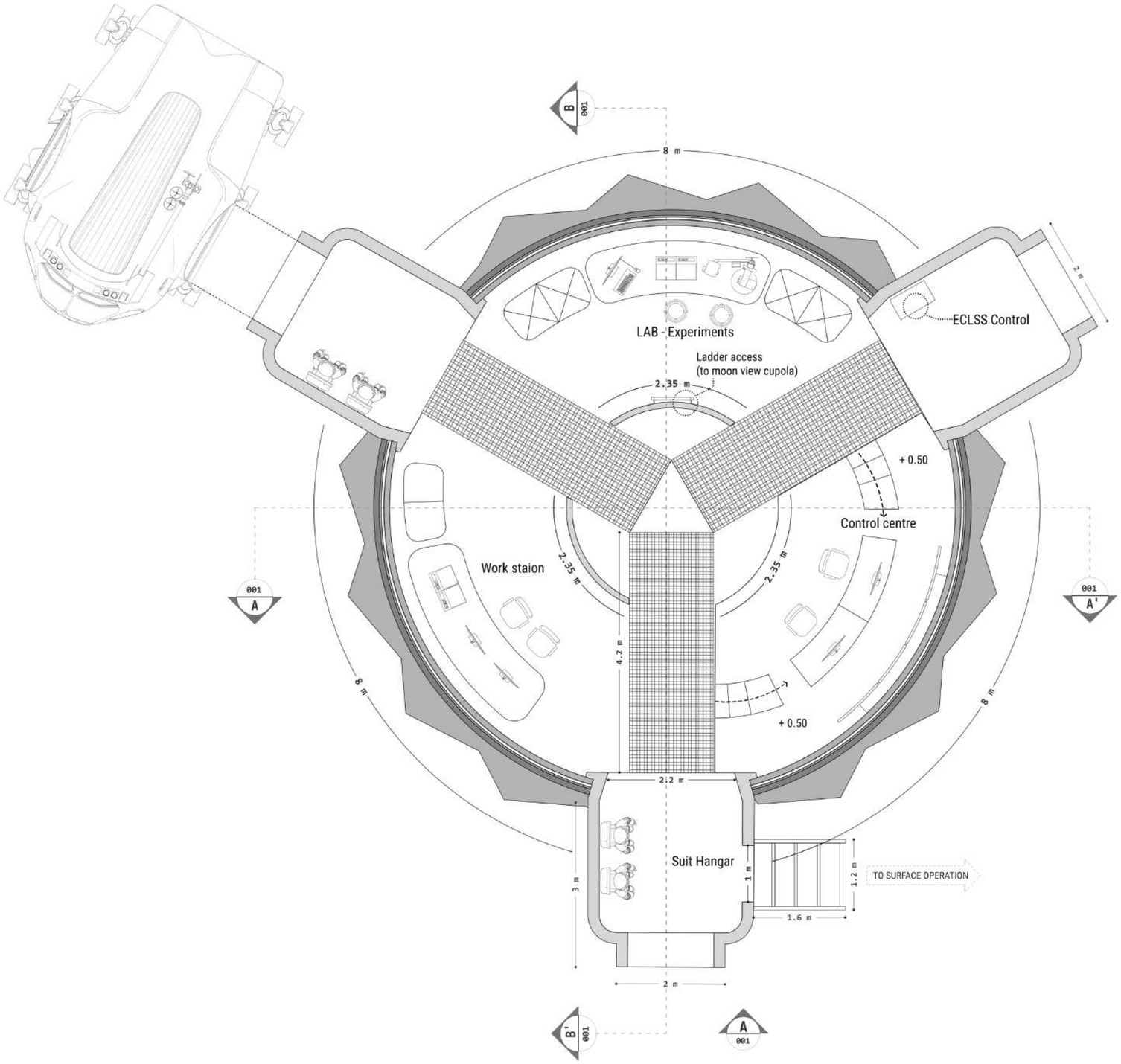
Labs | Plant habitat

The concluding stage entails establishing laboratories and experimental areas on the lunar surface through In-Situ Resource Utilization (ISRU). Lunar regolith is utilized for 3D printing modules directly on the surface with the assistance of robots. This initiative enhances our research capabilities and facilitates deeper exploration of the Moon. These facilities are strategically situated away from living modules, with communal spaces serving as a buffer zone. This advancement marks a significant step forward in our quest for lunar exploration and understanding.

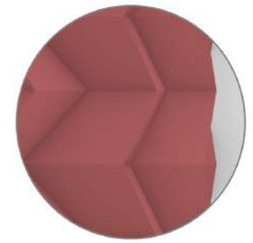




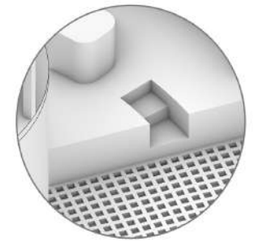
In the ongoing endeavor to establish a sustainable lunar colony, we find ourselves immersed in the third phase of development, marked by a profound emphasis on architectural innovation and adaptation. As we delve deeper into the complexities of lunar habitation, our focus shifts towards the creation of essential infrastructure and specialized facilities tailored to meet the unique challenges of life on the lunar surface. Within this phase, the architectural landscape of the lunar colony evolves to encompass laboratories, communal spaces, and experimental zones, each meticulously designed to harness the resources of the Moon and support the diverse needs of its inhabitants.



FLOOR PLAN - LAB



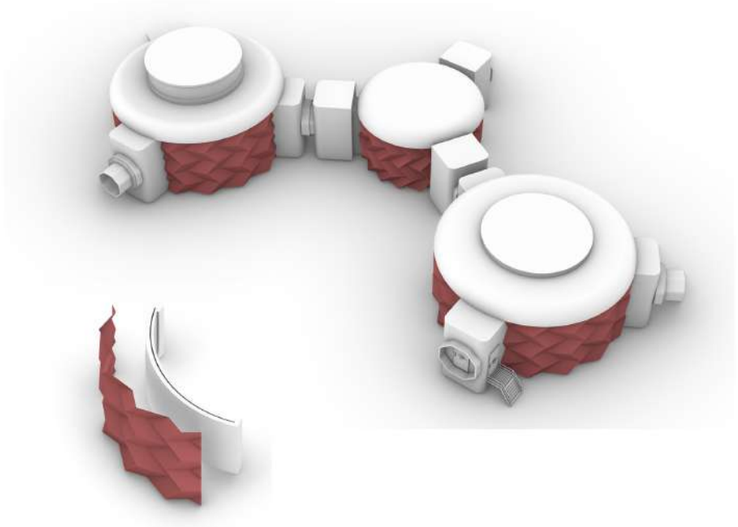
Miura tesselation pattern

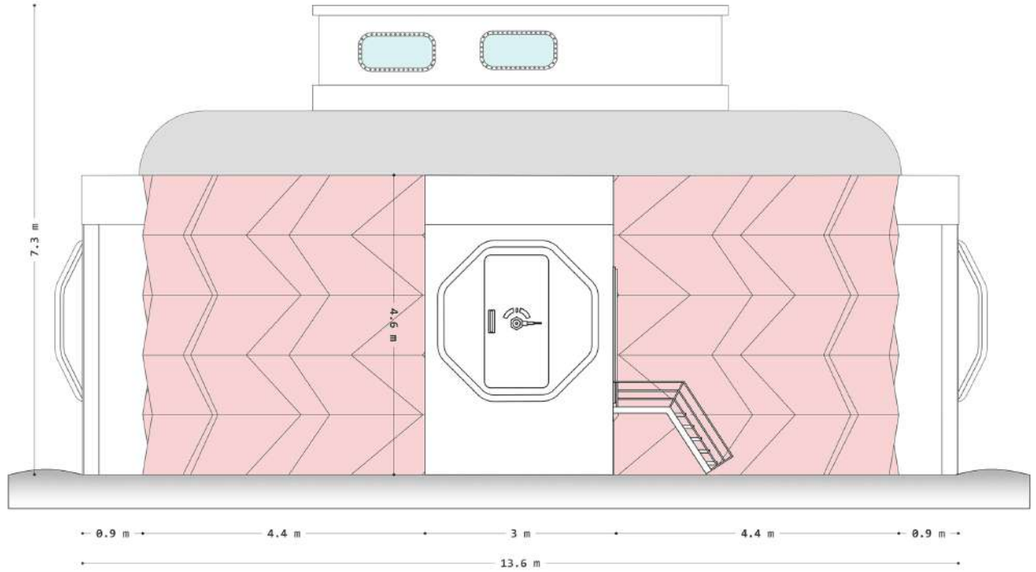


Aluminium grilled floor

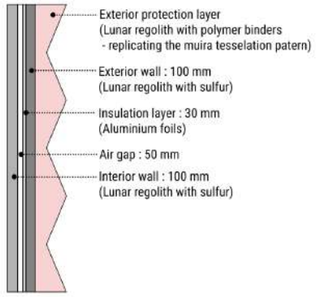


Moon view cupola

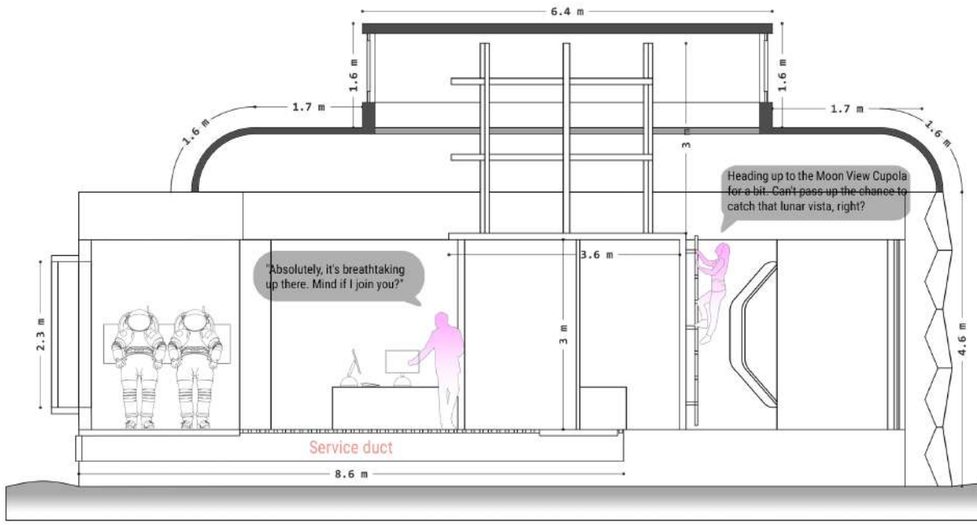




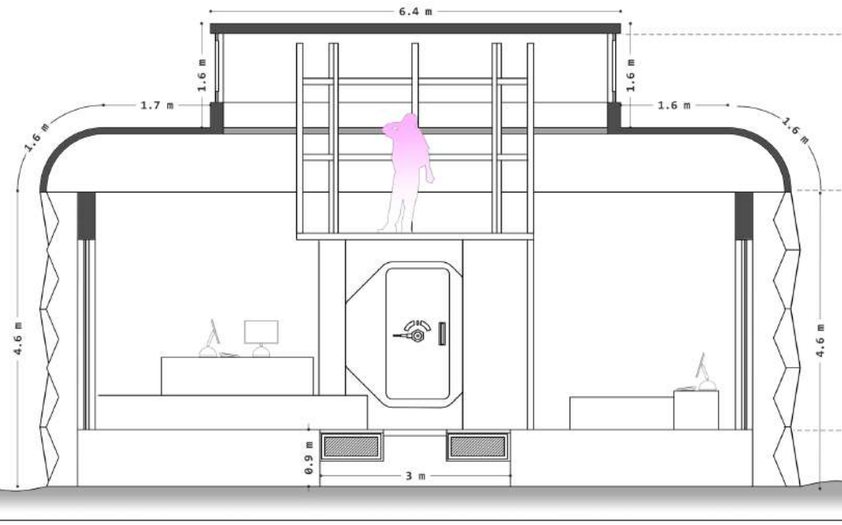
ELEVATION A
SCALE: 1:50 001



Wall layers



SECTION BB' 001
SCALE: 1:50



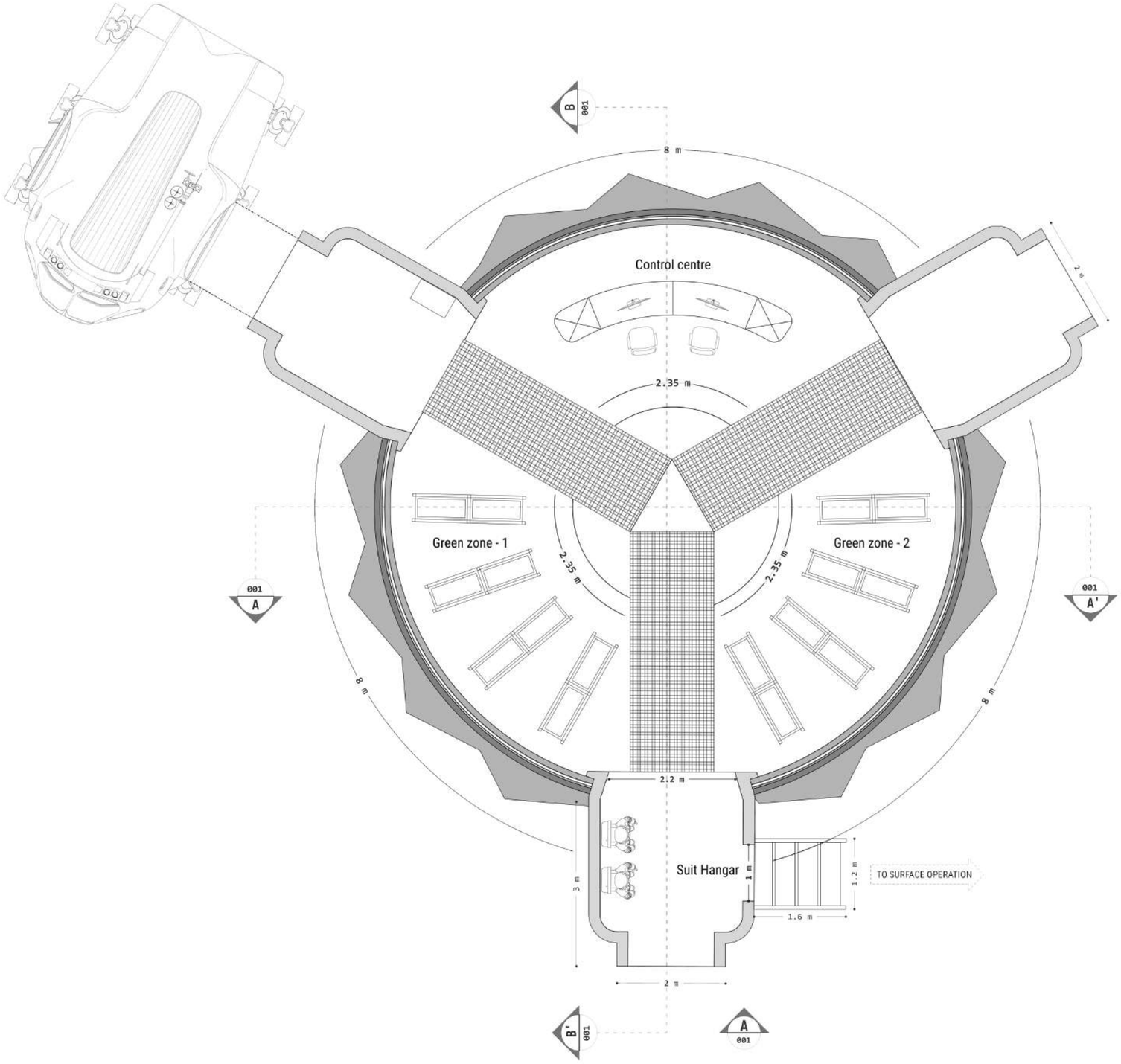
SECTION AA' 001
SCALE: 1:50

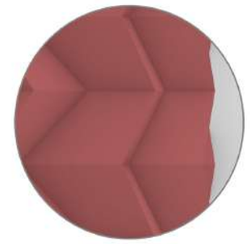
TOP OF ROOF
6.2 m from base

ROOF LEVEL
4.6 m from base

FLOOR LEVEL
0.9 m from base

BASE



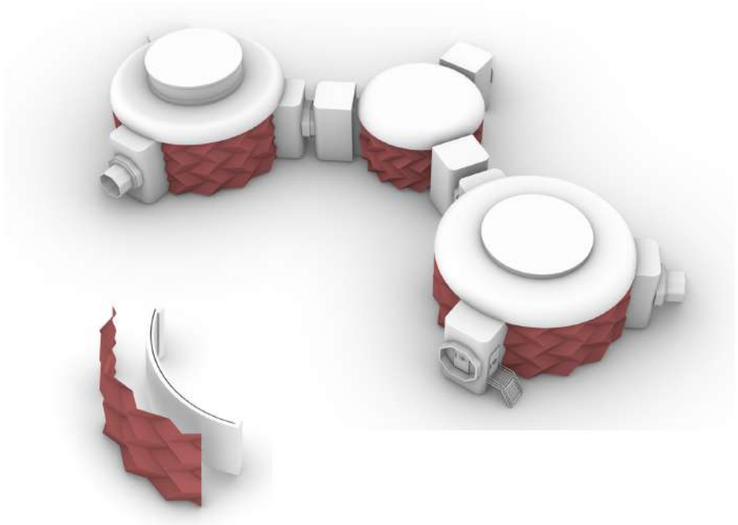


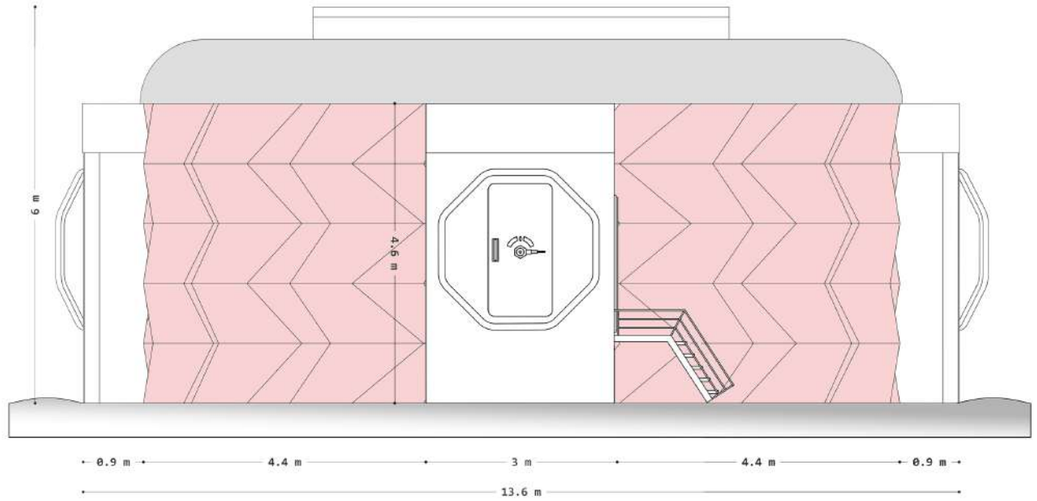
Miura tesselation pattern



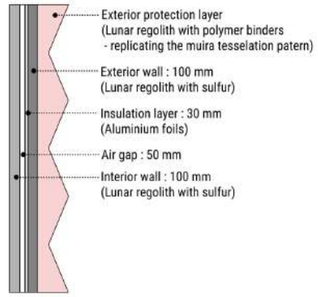
Plant racks

FLOOR PLAN - PLANT HABITAT

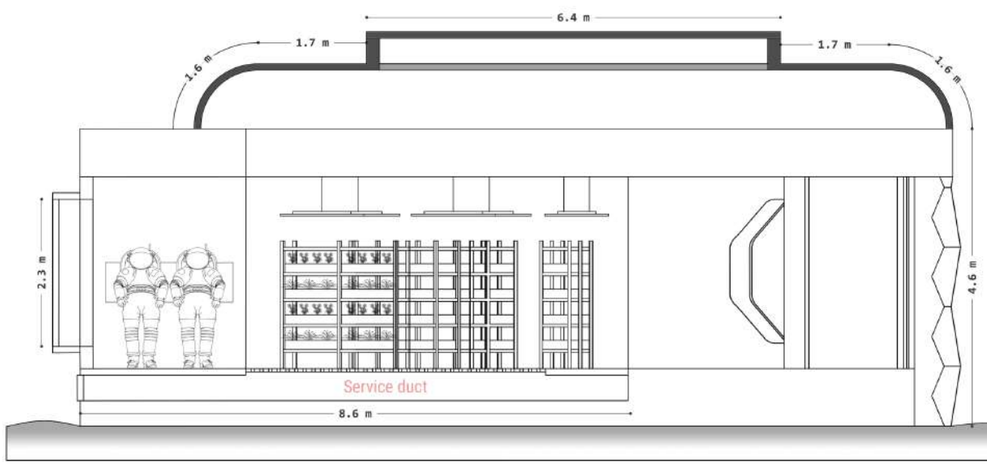




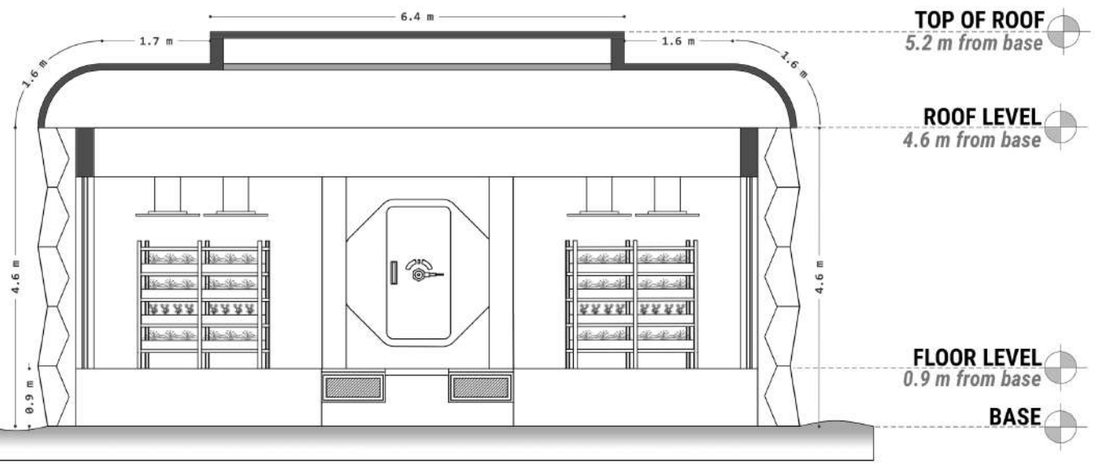
ELEVATION A 001
SCALE: 1:50



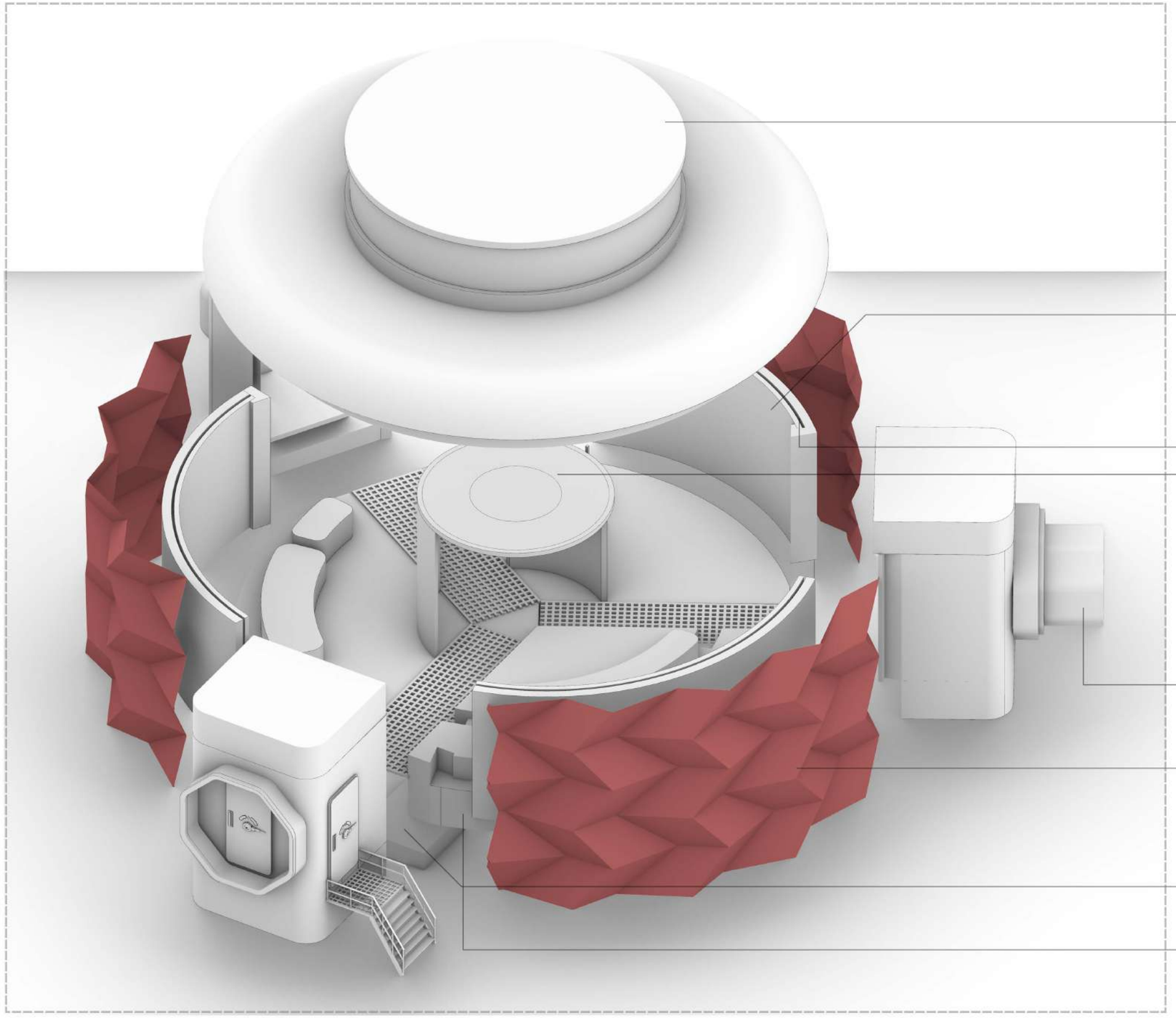
Wall layers



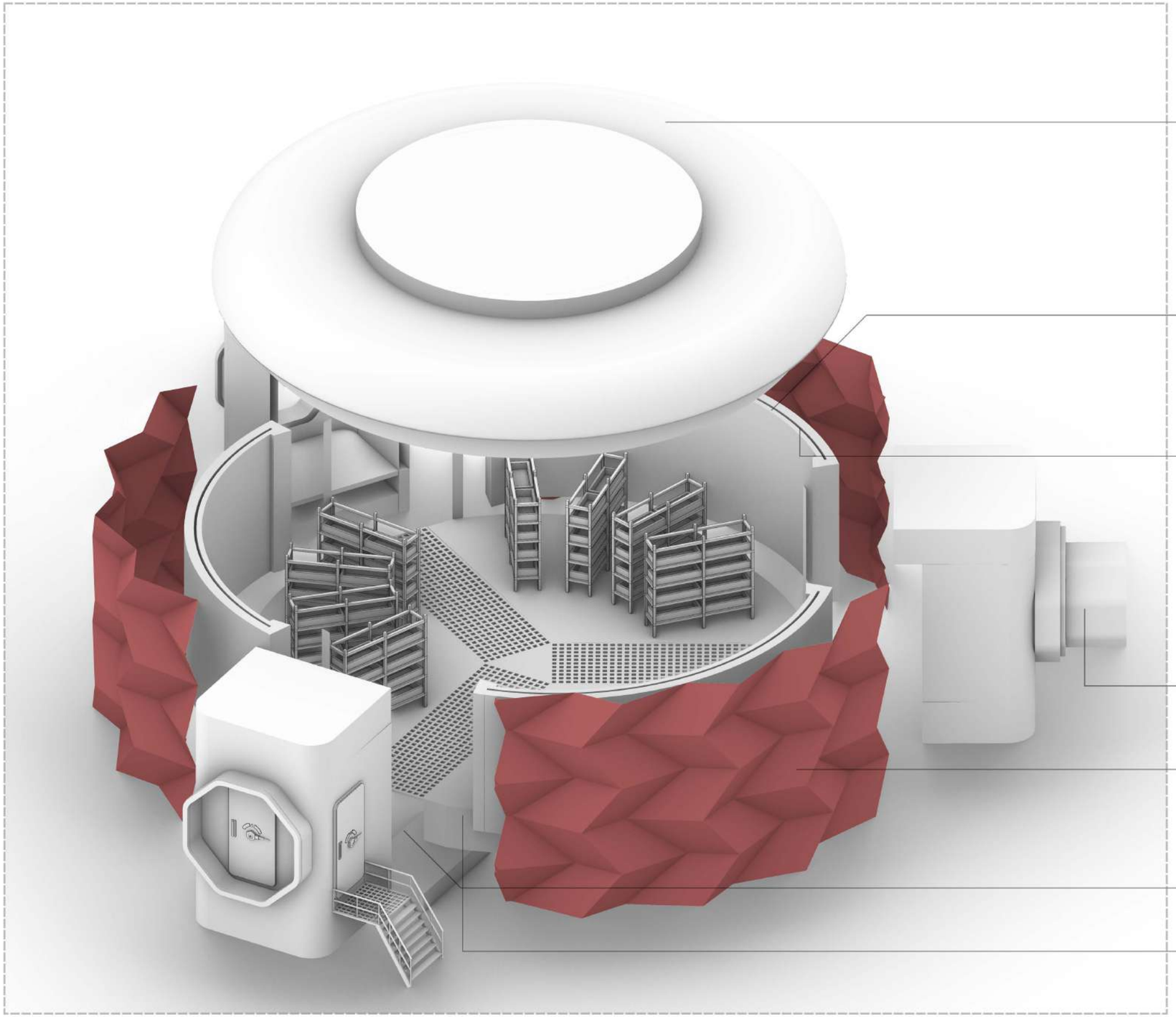
SECTION BB' 001
SCALE: 1:50



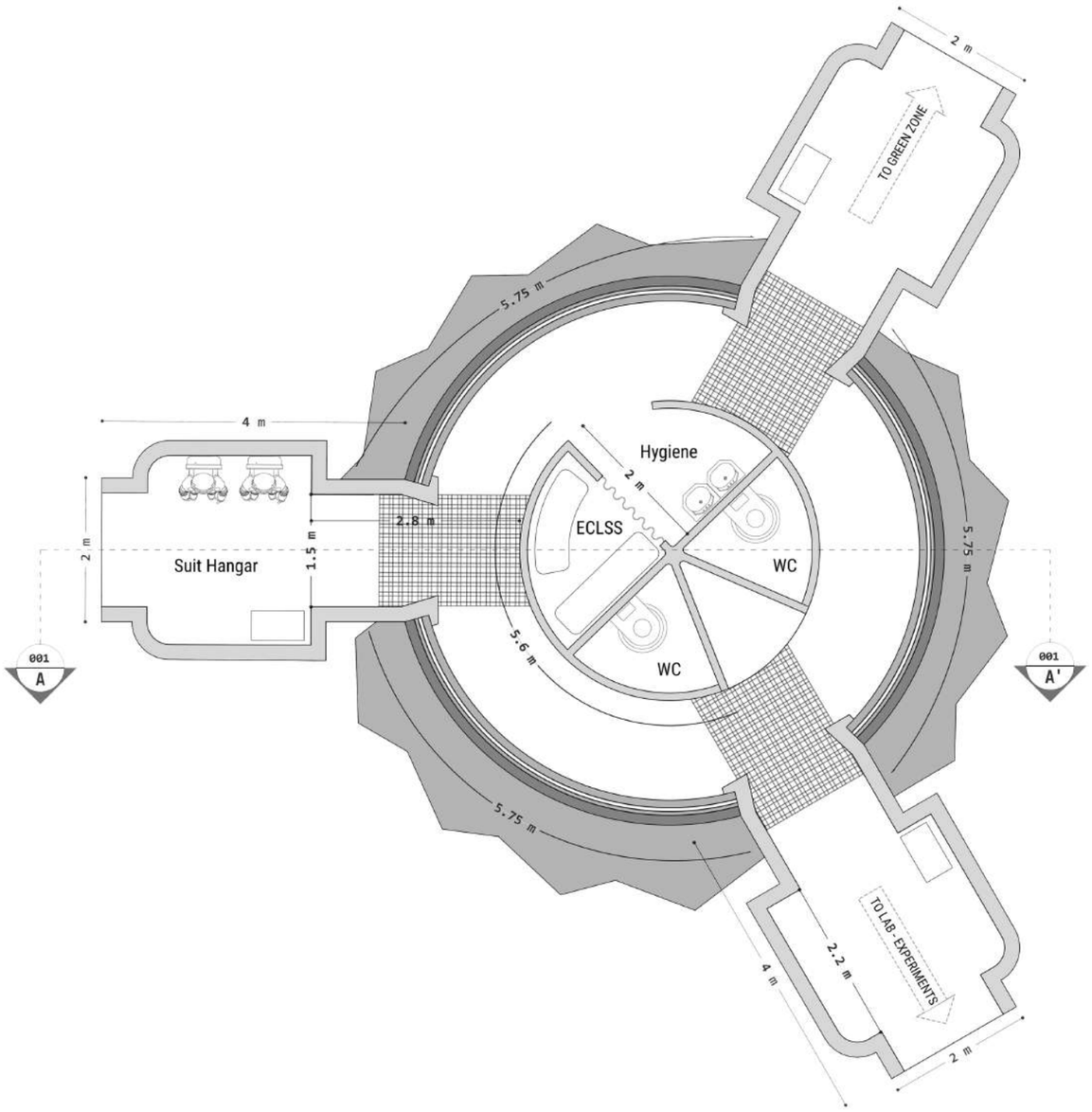
SECTION AA' 001
SCALE: 1:50



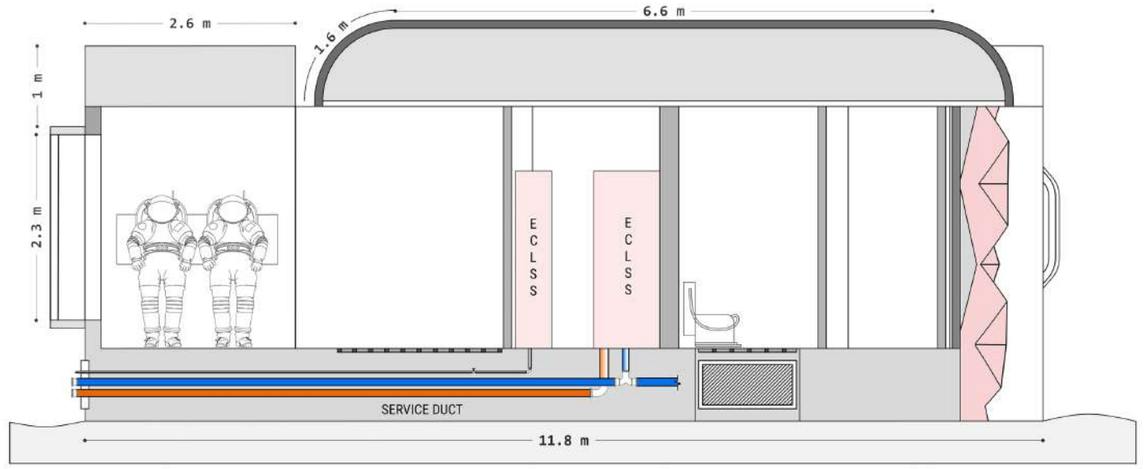
EXPLODED - LAB



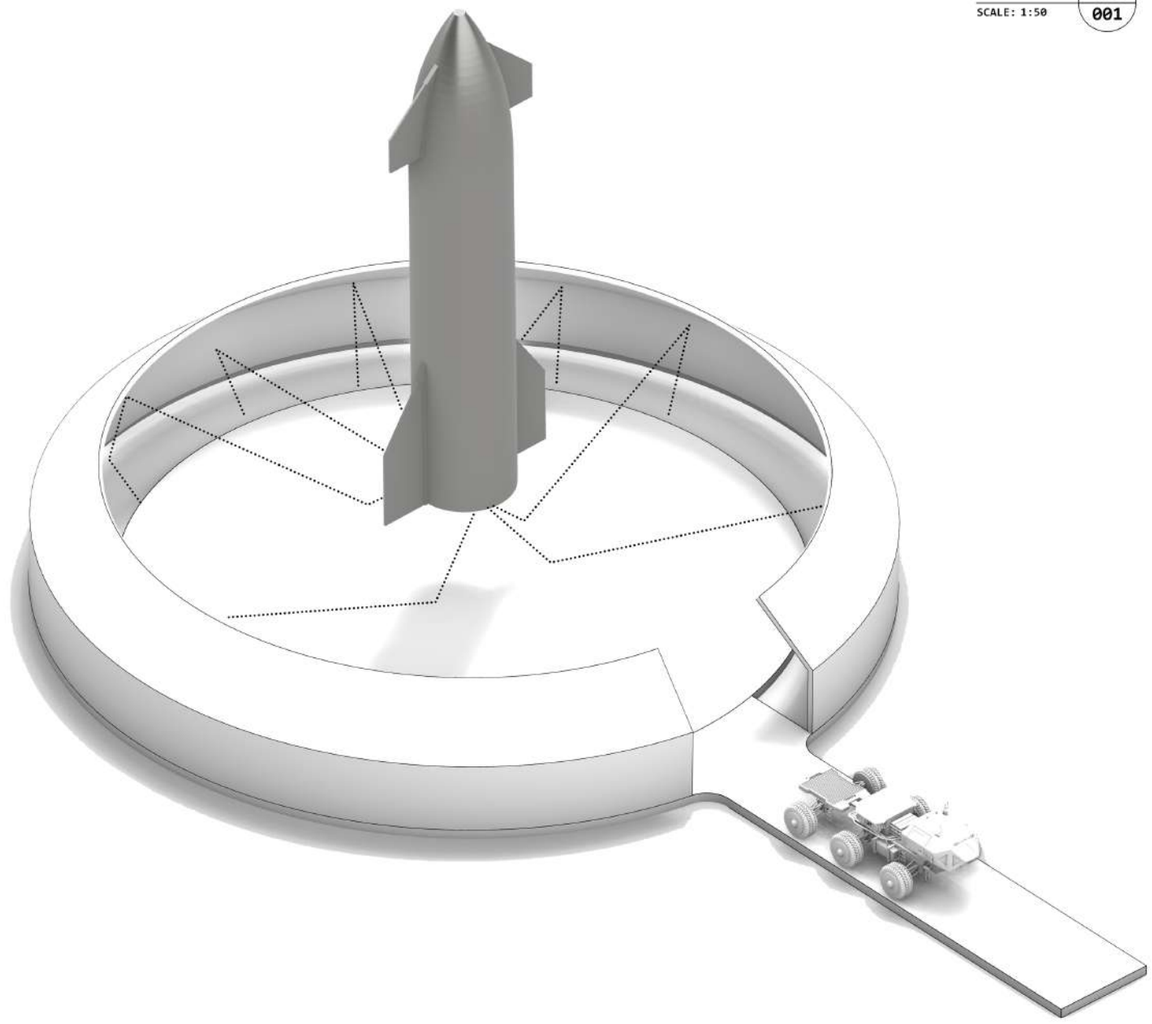
EXPLODED - PLANT HABITAT



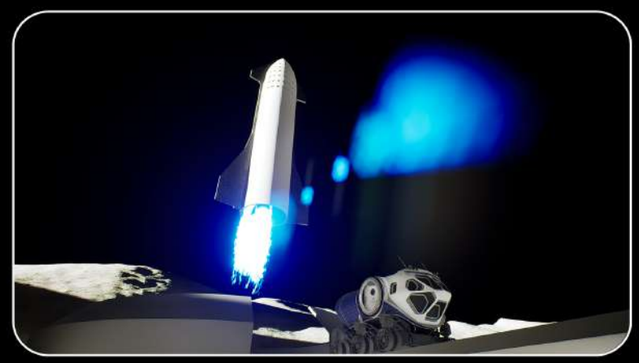
FLOOR PLAN - ECLSS MODULE



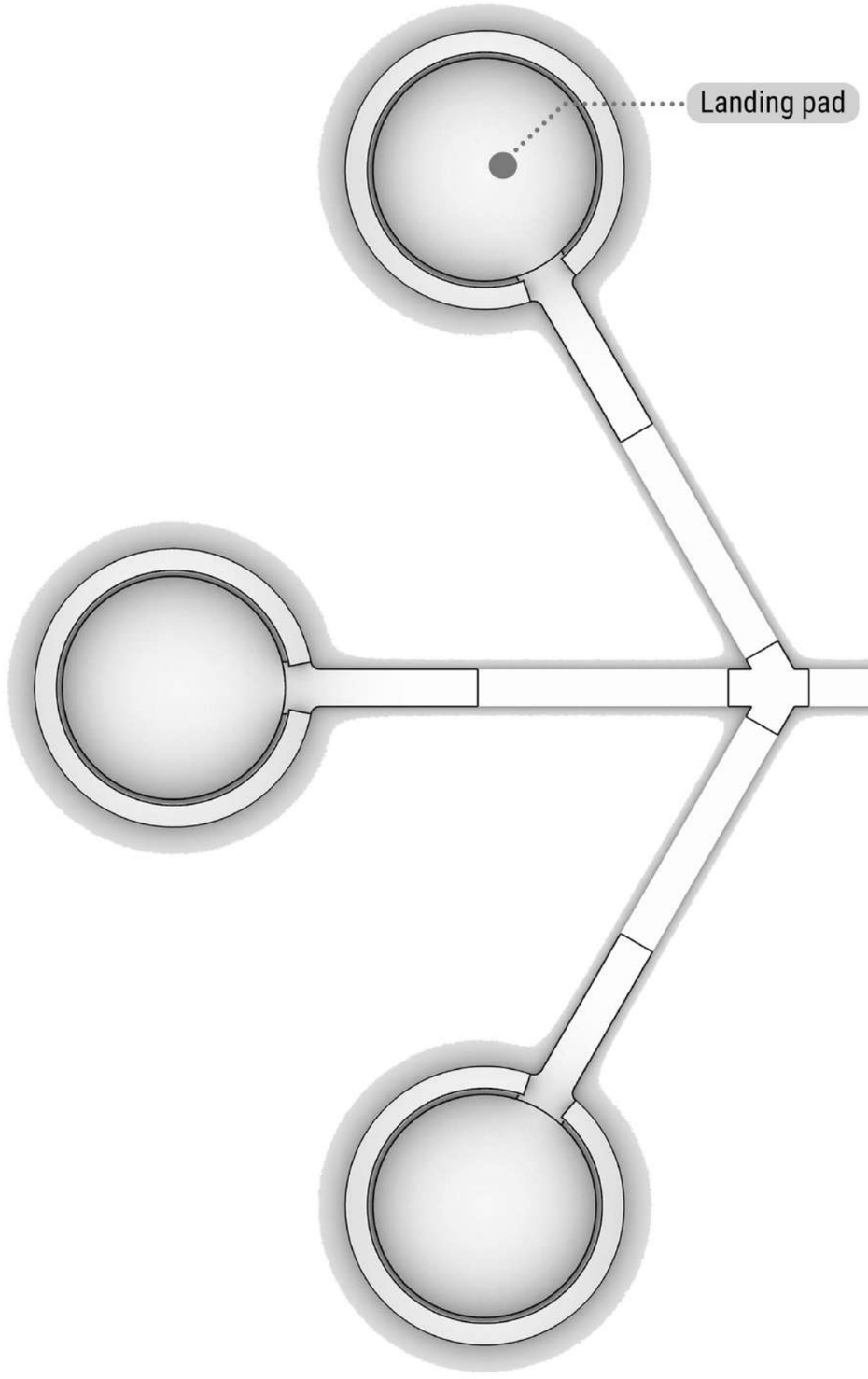
SECTION AA' 001
SCALE: 1:50

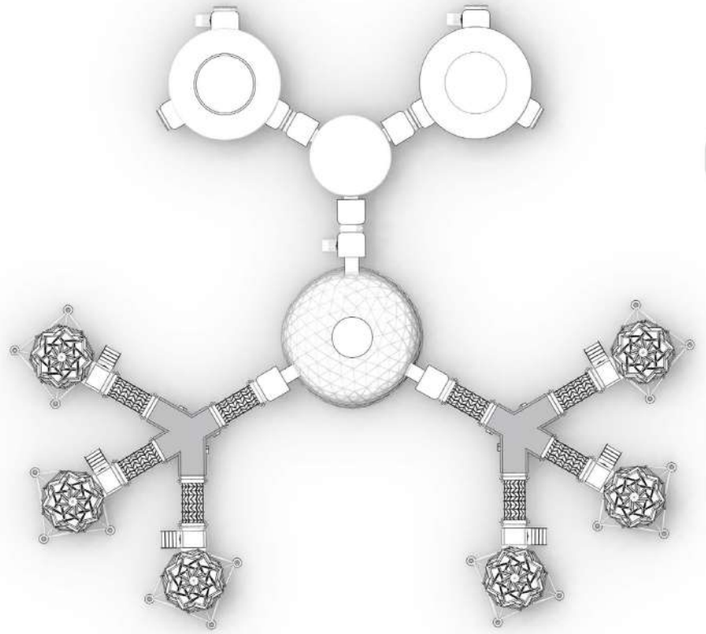




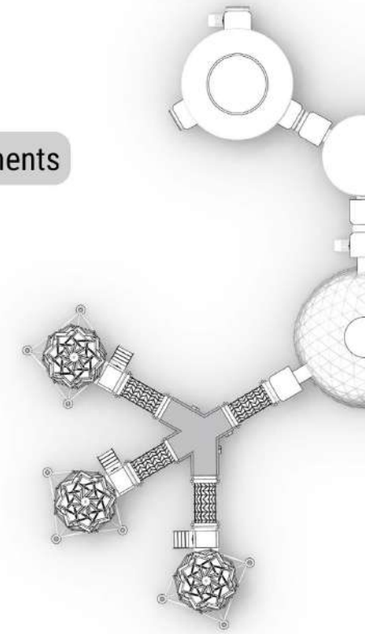


The spaceport prevents lunar debris from entering the habitable zones. A paved lunar surface aids rovers in transporting people and goods to and from the spaceport, ensuring efficient and safe lunar travel.

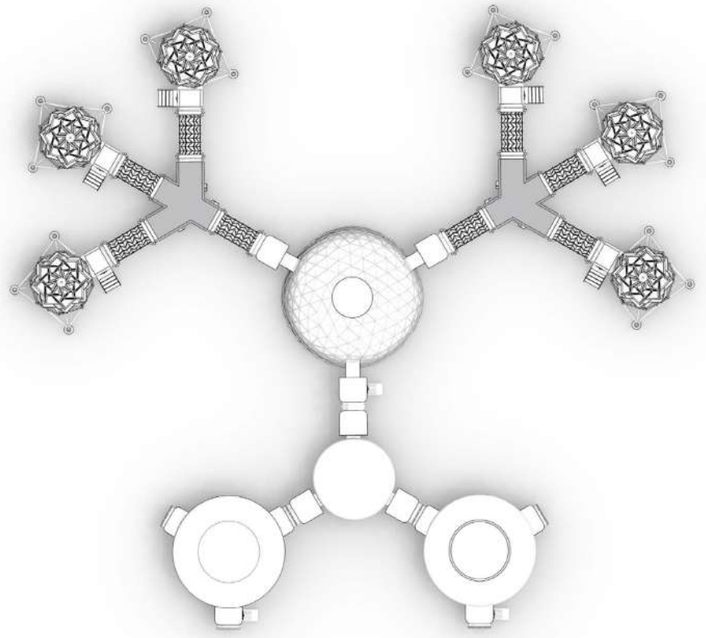


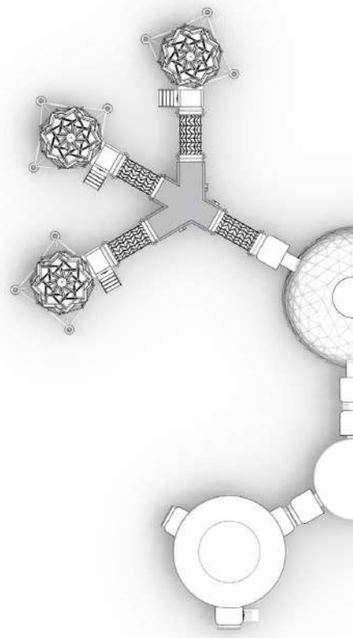


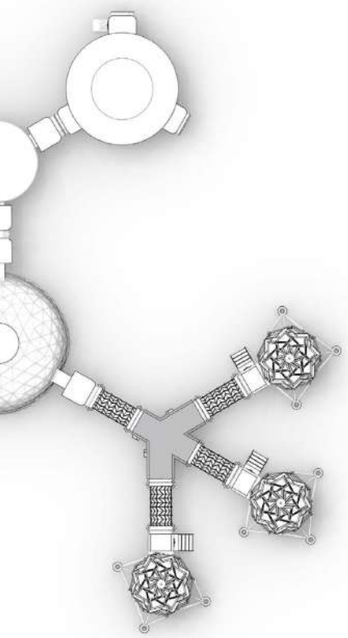
Cluster settlements

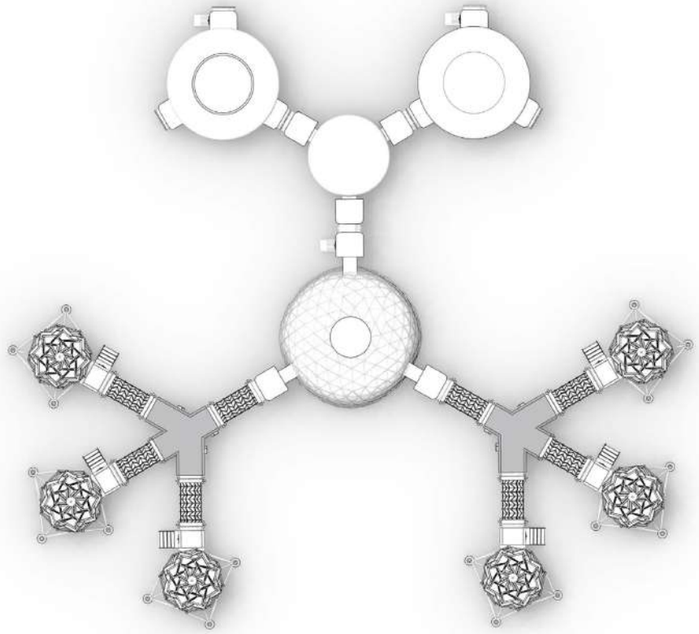


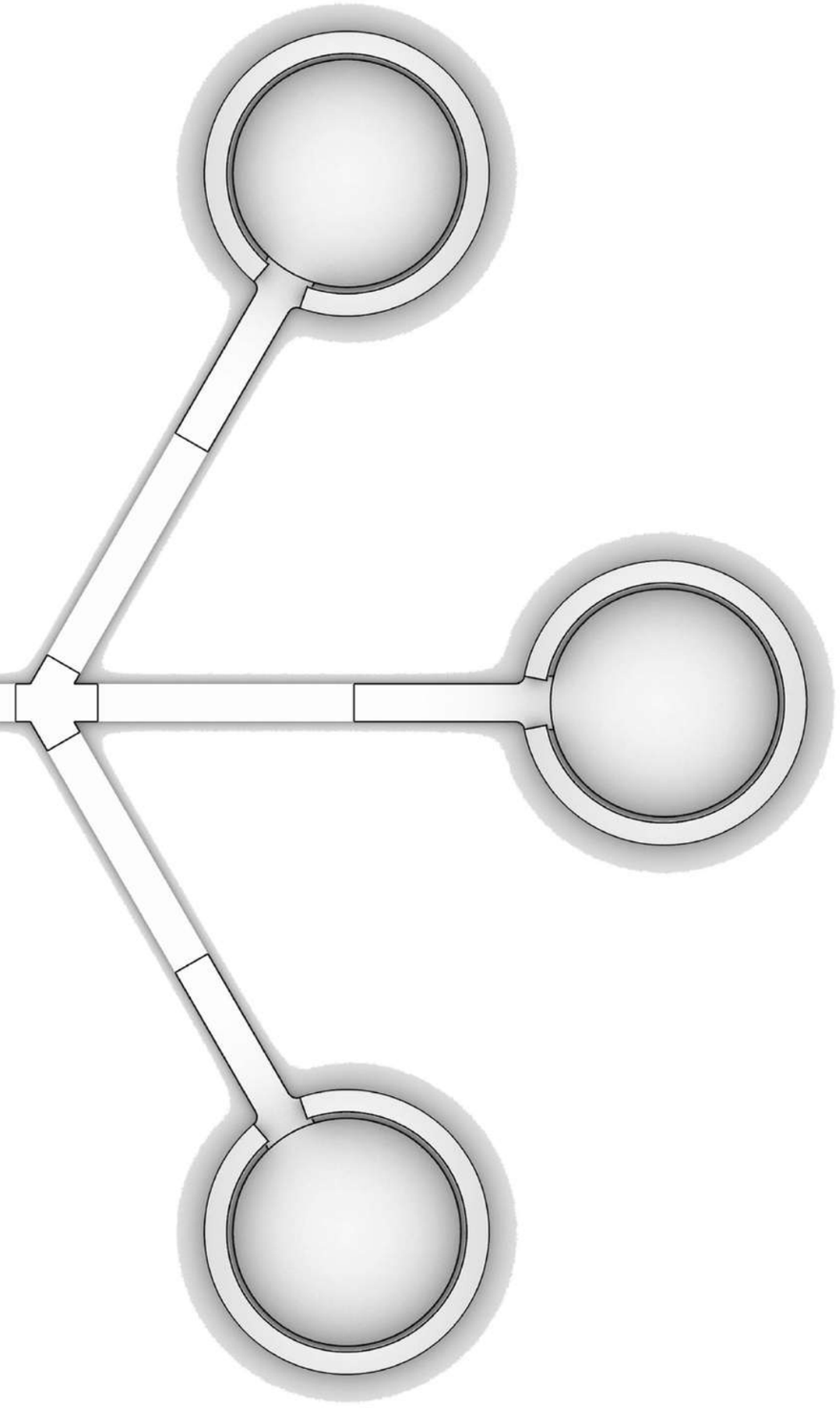
Rover tra



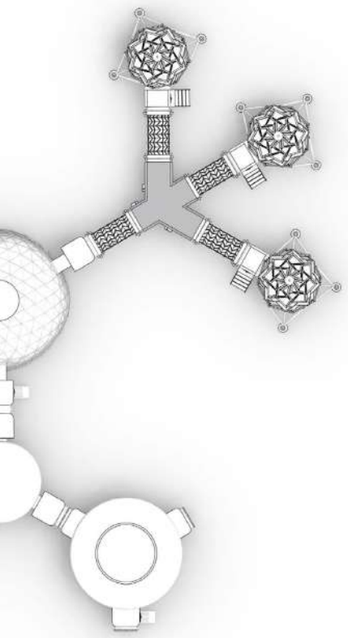


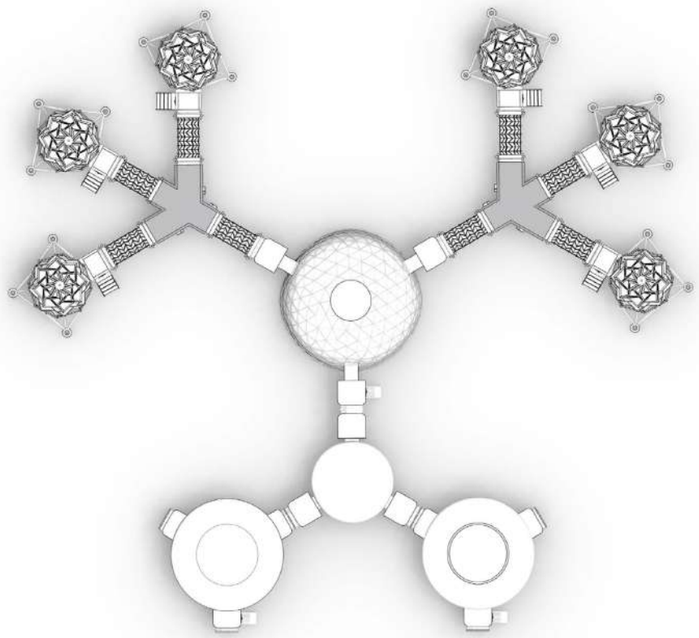


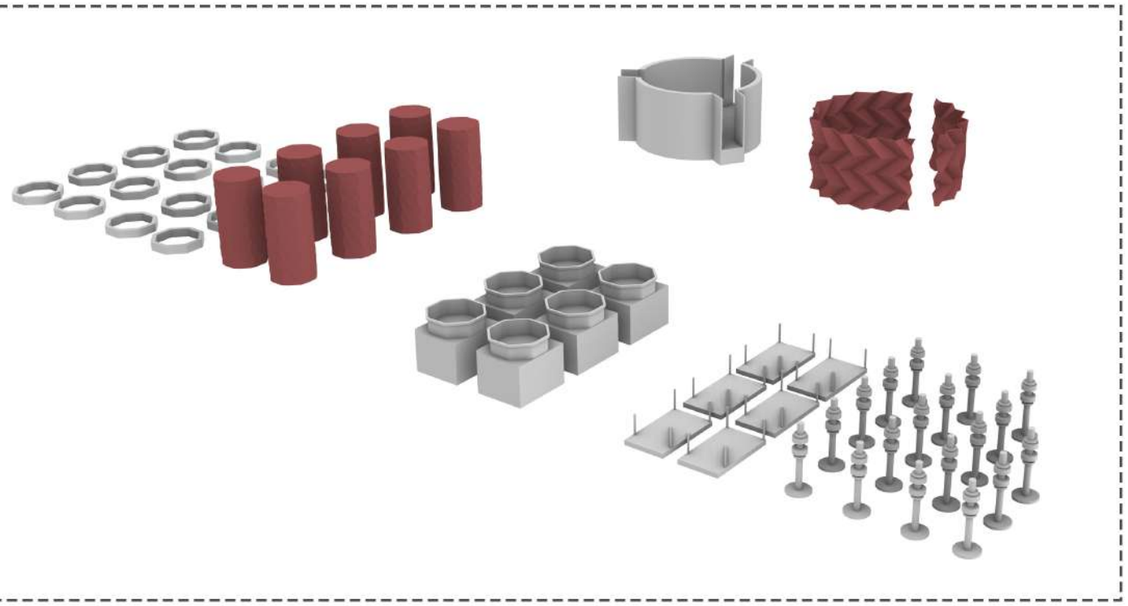




nsit zone



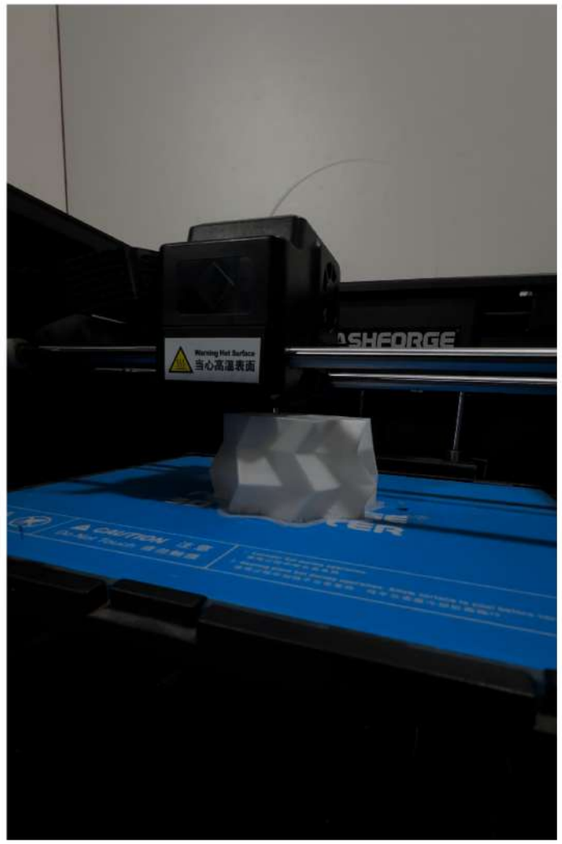


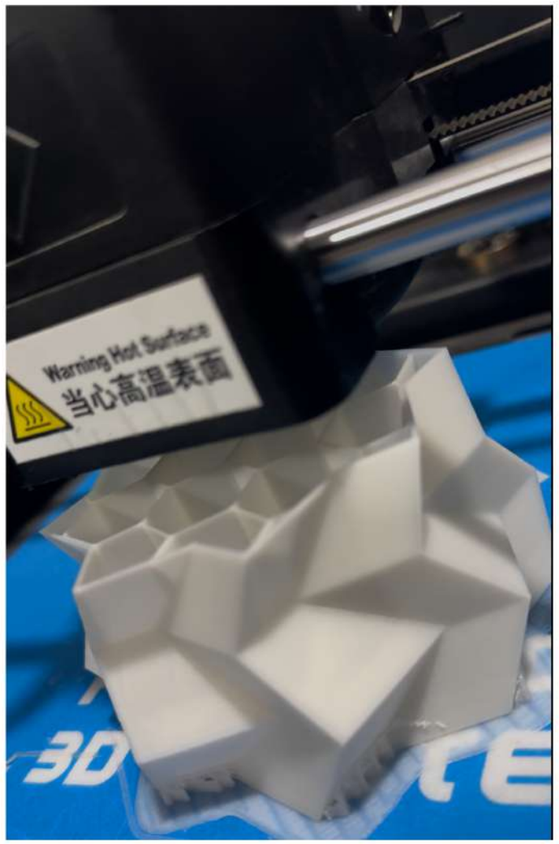


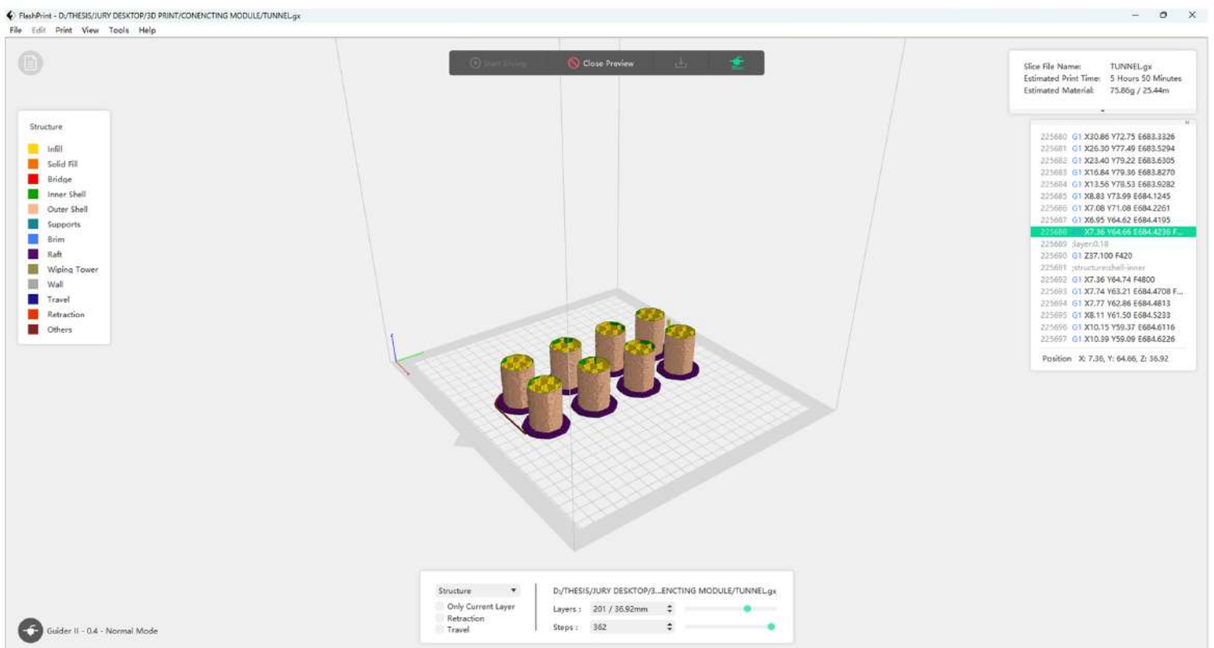
Additive manufacturing

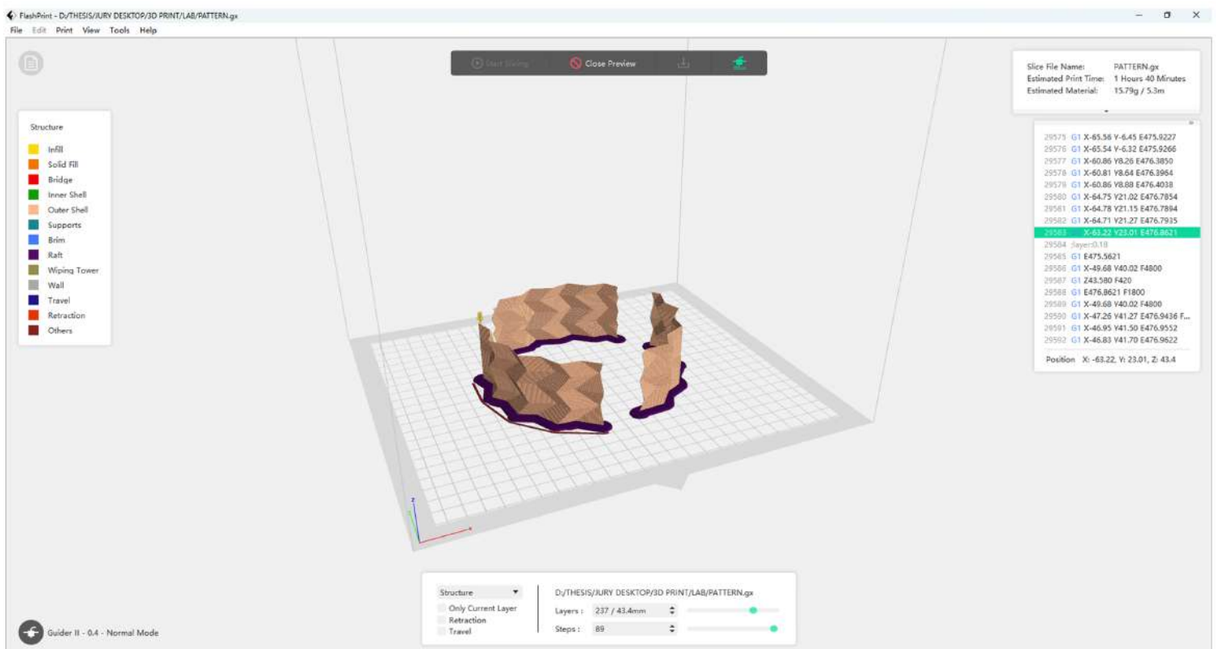
Digital modeling | Slicing | 3d printed model

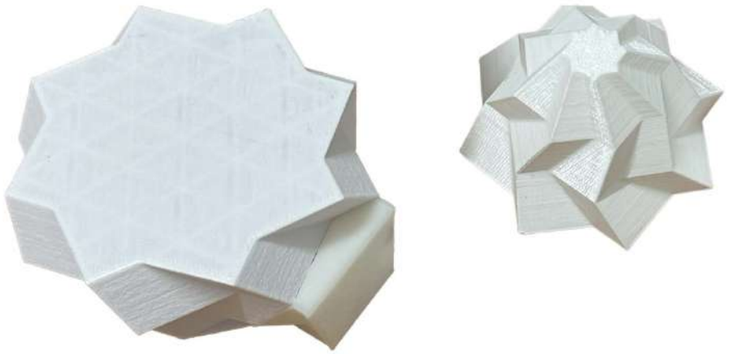
3D printing technology is being utilized to create detailed and accurate models of lunar bases. This innovative approach allows for the precise replication of intricate designs and structural elements that are crucial for habitation in the moon's harsh environment. By using advanced materials and techniques, the models can simulate the unique conditions and challenges of lunar construction, providing a tangible representation of the concepts. These 3D printed models not only enhance the visual and functional aspects of the thesis but also demonstrate the potential of additive manufacturing in space architecture, paving the way for future exploration and settlement on the moon.







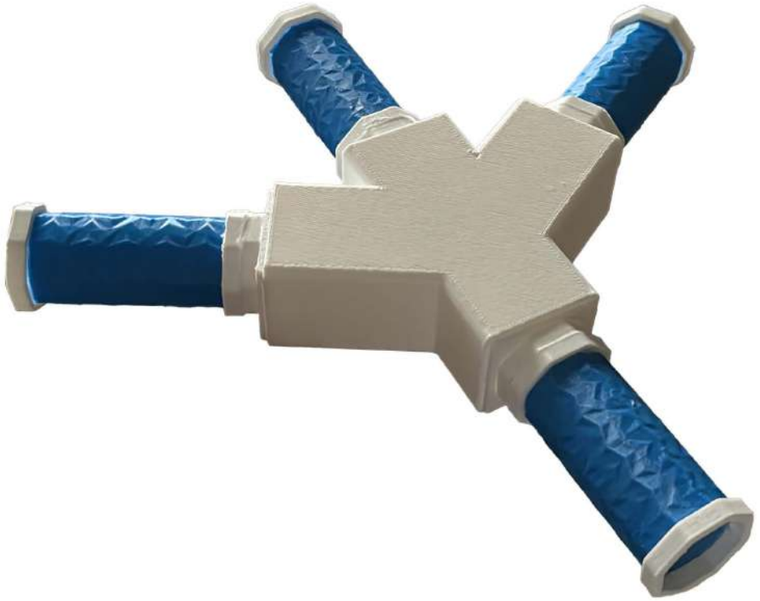
















thank**you.**